PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 6A

Financial Management

Short Notes

For every lesson, in syllabus order: the key idea, the formats and formulas, the answer frame, the common mistakes with their fixes, and the recap.

32 lessons in 9 chapters · phone edition · made 2026-10-20 from the PassBook lessons

The labs (step-through simulators) are not printed: open the lesson in the PassBook app for the interactive version.

Contents

1 Scope and Objectives of Financial Management 5

1.1 Objectives: profit maximisation, wealth maximisation and value creation 5

1.2 The finance function, the financial manager and the controller 12

1.3 Agency problem, financial environment, distress and insolvency 17

2 Financial Analysis and Planning: Ratio Analysis 21

2.1 Liquidity and capital structure ratios 21

2.2 Activity (turnover) ratios 32

2.3 Profitability and market ratios, and the DuPont analysis 41

2.4 Building financial statements from ratios; limitations of ratios 51

3 Types of Financing 60

3.1 Long-term sources: equity, preference, debt and retained earnings 60

3.2 Venture capital, lease financing, sale and lease back and other sources 65

3.3 Short-term sources of finance 70

3.4 Contemporary and international sources of finance 75

4 Cost of Capital 80

4.1 Cost of debt and of preference capital 80

4.2 Cost of equity and of retained earnings 89

4.3 WACC and the marginal cost of capital 96

5 Financing Decisions: Capital Structure 103

5.1 Capital structure theories: NI, NOI, traditional and Modigliani-Miller 103

5.2 EBIT-EPS analysis, indifference point and financial break-even 111

5.3 Capital structure planning, determinants and over- and under-capitalisation 119

6 Financing Decisions: Leverages 123

6.1 Operating, financial and combined leverage 123

7 Investment Decisions 133

7.1 Capital budgeting and estimating relevant cash flows 133

7.2 Payback, discounted payback and ARR 143

7.3 NPV, profitability index and equivalent annual cost 150

7.4 IRR, MIRR and the NPV-IRR conflict 155

7.5 Capital rationing 163

8 Dividend Decision 168

8.1 Forms of dividend, buy-back and determinants of dividend policy 168

8.2 Dividend theories: traditional, Walter, Gordon and Modigliani-Miller 174

9 Management of Working Capital 180

9.1 Working capital, the operating cycle and financing policy 180

9.2 Estimating working capital requirement 188

9.3 Management of receivables, credit policy, factoring and forfaiting 196

9.4 Treasury and cash management 207

9.5 Management of inventory 217

9.6 Management of payables 224

9.7 Financing of working capital and bank credit 232

CHAPTER 1

Scope and Objectives of Financial Management

1.1 Objectives: profit maximisation, wealth maximisation and value creation

Key idea

Profit maximisation asks "how much profit?"; wealth maximisation asks "how much are the owners' shares worth?", and so counts cash flows, their timing and their risk. In an uncertain, multi-period world, wealth (value) maximisation is the better objective of financial management.

FORMATS AND FORMULAS

Formula

Shareholders' wealth

$$\text{Wealth} = \text{PV of benefits} - \text{PV of costs}$$

PV = present value. Benefits and costs are cash flows, discounted for timing and risk (SM 1.14).

∑ **Formula**

Value of a firm

$$V = N \times MP \qquad \text{or} \qquad V = V_e + V_d$$

V = value of the firm; N = number of shares; MP = market price of a share; Ve = value of equity; Vd = value of debt (SM 1.15).

The comparison ICAI prints

Reproduce this table from memory: ICAI's suggested answers copy it row for row (SM 1.17).

Point

Objective

Profit maximisation

Large amount of profits

Wealth maximisation

Highest market value of shares

Point

Advantages

Profit maximisation

(i) Easy to calculate profits; (ii) easy to determine the link between financial decisions and profits

Wealth maximisation

(i) Emphasises the long-term gains; (ii) recognises risk or uncertainty; (iii) recognises the timing of returns; (iv) considers shareholders' return

Point	Disadvantages
Profit maximisation	(i) Emphasises the short-term gains; (ii) ignores risk or uncertainty; (iii) ignores the timing of returns; (iv) requires immediate resources
Wealth maximisation	(i) Offers no clear relationship between financial decisions and share price; (ii) can lead to management anxiety and frustration

Definition

Shareholder value maximisation model

The model which holds that the primary goal of the firm is to maximise its market value, and implies that business decisions should seek to increase the net present value of the economic profits of the firm. (SM 1.14)

ANSWER FRAME

"Explain the limitations of profit maximisation" (2 or 4 marks)

1. One line of context: profit maximisation cannot be the sole objective; it is at best a limited objective.
2. The four limitations in ICAI's order, each with its one-line reason: vague (short or long term, total or rate); risk ignored (higher risk, higher possible profit, risky proposals accepted); timing ignored (proposal A's

returns 10 years later); too narrow (social considerations, short-sighted policy).

"Distinguish between profit maximisation and wealth maximisation"

Point	Profit maximisation	Wealth maximisation
Objective	Large amount of profits	Highest market value of shares
Measure	Accounting profit	Cash flows, discounted (PV of benefits – PV of costs)
Risk	Ignores risk or uncertainty	Recognises risk or uncertainty
Timing	Ignores the timing of returns	Recognises the timing of returns
Time horizon	Emphasises short-term gains	Emphasises long-term gains
Sustainability	Low: not sustainable in the long term	High: critical for the very existence of the business

"Comment" on a statement or a case

Verdict first ("The statement is true / not correct because..."), then the relevant limitations or advantages from the Study Material, then the facts of the case applied

to them, then the conclusion: which objective the decision serves, and which ICAI prefers (wealth maximisation, in an uncertain multi-period world).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: "wealth maximisation means making the owners rich at everyone else's cost"**
   Fix: the Study Material says the opposite: the wealth maximisation objective is generally in accord with the interests of owners, employees, creditors and society, and so may be consistent with management's objective of survival (SM 1.16). Its price is a report card that counts the long term.

2. **Writing the comparison table's disadvantages as the four limitations**
   Fix: answer the list the question names; when the question says "disadvantages", use the table.

3. **Calling the larger total profit the larger wealth**
   Fix: wealth is present value of benefits less present value of costs; a larger profit that comes later or with more risk can be worth less.

4. **Saying wealth maximisation "always" differs from profit maximisation**
   Fix: where the time period is short and the degree of uncertainty is not great, the two amount to essentially the same (SM 1.16). They part company when time and risk matter.

RECAP

- Profit maximisation is at best a limited objective: vague, ignores risk, ignores the time pattern of returns, too narrow (in that order).
- Wealth maximisation measures success by the market value of the shares; wealth = PV of benefits − PV of costs.
- To measure wealth: cash flow approach (not accounting profit), cost-benefit analysis, time value of money.
- $V = N \times MP$, or $V = Ve + Vd$. The market price is the "report card" of the firm (Van Horne).
- Value creation: decisions that earn sustainable returns above the cost of capital (ICAI's MTP answer); the SM treats it with wealth maximisation.
- Wealth maximisation stays the primary goal among growth, market share, leadership, employee welfare, customer satisfaction and community goals.
- In an uncertain, multi-period world wealth maximisation is better; with a short period and little uncertainty the two are essentially the same.

- Comparison table: profit (easy to calculate; short-term, ignores risk and timing, requires immediate resources) against wealth (long-term, risk, timing, shareholders' return; no clear link to share price, management anxiety). Kesar's record quarter cost him 6%.

1.2 The finance function, the financial manager and the controller

Key idea

Financial management is the efficient acquisition (financing) and allocation (investment) of funds, with the dividend decision on top: three inter-related long-term decisions plus working capital management. The finance executive who takes them has grown from a fund-raiser for big events into the CEO's strategic partner.

ANSWER FRAME

"Discuss the role (responsibilities) of the finance executive / CFO"

Open with one sentence: the finance executive plays an important role in the company's goals, policies and financial success. Then the five responsibilities, each with its ICAI gloss: financial analysis and planning (size and rate of growth); investment decisions (efficient allocation of funds to specific assets); financing and capital structure decisions (composition of liabilities); management of financial resources (working capital); risk management (protecting assets). If the question says "in today's world", add the line on the strategic business partner of the CEO and the "now does" list.

"Explain the inter-relationship between investment, financing and dividend decisions"

1. The three decisions; inter-related because their objective is the same, maximisation of shareholders' wealth; consider their joint impact on the share price and solve them jointly.
2. Investment: capital budgeting; accept only if the return at least meets the cost of financing.
3. Financing: balance of long-term and short-term funds; mix of loan and owners' funds; the optimum mix maximises return to equity.
4. Dividend: payout against retention; retained earnings used for internal financing deprive shareholders of dividends; optimal payout maximises wealth.
5. Conclusion: take them jointly, keeping in view their joint effect on shareholders' wealth.

"Distinguish the treasurer's and the controller's functions"

Treasurer	Controller
Cash manager	Financial accounting manager
Credit manager	Cost accounting manager
Capital budgeting manager	Tax manager
Fund raising manager	Data processing manager
Portfolio manager	Internal auditor

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the retained profit that was "only a dividend decision"**

   Fix: the amount retained to support growth is also a financing decision (SM 1.9). Say both.

2. **Answering "role of the financial controller" with the organisation chart**

   Fix: SM 1.11's financial controller is the head of finance: once limited to procuring funds for major events, now taking the investment, financing and dividend decisions, maximising returns in relation to the given risk and monitoring the flow of funds. The chart's "Controller" is a different post.

3. **Calling equity "the cheapest source because there is no interest"**

   Fix: equity is the best source from the **risk** point of view (no repayment except in liquidation) but usually the most **expensive**: shareholders expect more than the interest rate, and dividends are not tax-deductible (SM 1.5). Debentures are cheaper but riskier.

4. **Listing the CFO's old jobs as the "role today"**
   Fix: the "now does" list adds managing M&As, profitability and pricing analysis, outsourcing decisions, overseeing IT and HR, strategic planning, regulatory compliance and risk management, on top of budgeting and forecasting.

RECAP

- Financial management is the efficient acquisition (financing) and allocation (investment) of funds, with the objective of making profit (dividend) for owners; mainly to maximise shareholders' wealth.
- Two aspects: procurement of funds (each source differs in risk, cost and control) and effective utilisation (fixed assets via capital budgeting, and working capital without too much blocked in it).
- V = f(I, F, D): investment (selection of assets), financing (optimum finance, sound capital structure, profit versus cash, risk, hedging) and dividend (payout and retention, the retention being also a financing decision); short-term: working capital management.
- The three decisions are inter-related through one objective, shareholders' wealth, and must be taken jointly.
- Scope (Ezra Solomon): size and growth; composition of assets; mix of financing; analysis, planning and control.

- Evolution: traditional (occasional events), transitional (day-to-day problems), modern (analysis and theories).
- Financial controller: once procurement for major events, now the three decisions; maximise returns for the given risk. Finance executive: planning, investment, financing and capital structure, management of financial resources, risk management.
- Under the CFO: treasurer (cash, credit, capital budgeting, fund raising, portfolio) and controller (financial accounting, cost accounting, tax, data processing, internal audit); today's CFO is the CEO's strategic partner. The matchbox has been retired.

1.3 Agency problem, financial environment, distress and insolvency

Key idea

The agency problem is the chance that managers place personal goals ahead of the owners' goals; the extra cost shareholders bear to monitor and control them is the agency cost. Financial distress is cash inflows too small for current obligations; insolvency, the inability to repay debts, is the result of continuous distress.

ANSWER FRAME

"Explain the agency problem and agency cost; how can it be addressed?" (4 marks)

1. **Separation**: owners participate in management in a proprietorship or partnership; in a company they do not, so owners and managers are separate.
2. **Principal–agent relationship**: managers may maximise salary and perks; agency problem = the chances that managers place personal goals ahead of the owners' goal.
3. **Agency cost**: the additional cost borne by shareholders to monitor and control managers; four types: monitoring, bonding, opportunity, structuring.
4. **Addressing it**: debt lenders, negative covenants (no borrowing beyond a point); shareholders, align interests: compensation linked to profit and long-term

objectives; employee schemes linked to the share price; effective monitoring.

"Write a short note on financial distress and insolvency"

Factors to manage (price, demand, input prices, the proportion of debt; creditors' pressure) → definition of distress (cash inflows inadequate to meet all current obligations) → if continued, assets sold, often at a lower price → revenue inadequate to revive → definition of insolvency (inability to repay various debts, a result of continuous distress).

"Differentiate financial management and financial accounting"

Two heads, as in the comparison above: **treatment of funds** (accrual against cash flow, with the uncollectible-receivables example and the solvency reason) and **decision-making** (collecting and presenting data against planning, controlling and decision-making), ending with "financial management begins where accounting ends".

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: "profitable, so not in distress"**
   Fix: distress is about cash inflows against current obligations; accounting profit is on the accrual basis and can coexist with a cash shortfall.

2. **Using "distress" and "insolvency" as synonyms**
   Fix: distress is a cash shortfall against current obligations; insolvency is the inability to repay debts, the result of distress that continues and that revenue cannot reverse. A firm can come out of distress; insolvency is where distress ends if it does not.

3. **Saying the agency cost is borne by the managers**
   Fix: ICAI's definition: the additional cost borne by the **shareholders** to monitor and control the managers (SM 1.24).

4. **Forgetting the lender**
   Fix: the agency problem arises when managers' interests are not aligned to those of the debt lenders as well as the equity investors; lenders use negative covenants.

RECAP

- Agency problem: the chances that managers place personal goals ahead of the owners' goal; it arises from the separation of ownership and management in companies.
- Agency cost: the additional cost borne by shareholders to monitor and control managers; four types: monitoring, bonding, opportunity, structuring.
- Lenders' agency problem: negative covenants (managers cannot borrow beyond a point).

- Shareholders: link pay to profit and long-term objectives; employee schemes tied to the share price; effective monitoring.
- Financial distress: cash inflows inadequate to meet all current obligations; prolonged, it forces asset sales, often at lower prices.
- Insolvency: inability to repay various debts, the result of continuous distress when revenue cannot revive the situation.
- Accounting (accrual) is the primary input; financial management (cash flows; planning, control and decisions) begins where accounting ends; marketing, production and quantitative methods support, economics shows the world outside.
- The driver with the fuel card is cheaper to watch than to replace, but never free.

CHAPTER 2

Financial Analysis and Planning: Ratio Analysis

2.1 Liquidity and capital structure ratios

Key idea

A single figure says little; a figure compared with another says a lot. Liquidity ratios test whether short-term liabilities can be paid from assets that turn into cash within the year; capital structure ratios show how much of the business owners fund and how much lenders; coverage ratios show how many times earnings cover interest, instalments and preference dividend.

FORMATS AND FORMULAS

Reproduce the two lists and every formula from memory; ICAI's lists are what its answers count as current.

Current assets (SM 3.5)	Current liabilities (SM 3.6)
Inventories	Creditors for goods and services
Sundry debtors	Short-term loans
Cash and bank balances	Bank overdraft
Receivables / accruals	Cash credit
Loans and advances	Outstanding expenses
Disposable investments	Provision for taxation
Any other current assets	Proposed dividend
	Unclaimed dividend
	Any other current liabilities

Definition

Ratio (SM 3.3)

"The indicated quotient of two mathematical expressions and as the relationship between two or more things." A financial or accounting ratio is a mathematical expression of the relationship between two accounting figures.

∑ **Formula**

Current ratio and quick (acid-test) ratio

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Quick assets}}{\text{Current liabilities}}$$

Quick assets = current assets − inventories − prepaid expenses. ICAI's norms: 2:1 and 1:1 (SM 3.38).

∑ **Formula**

Cash ratio (absolute liquidity ratio)

$$\text{Cash ratio} = \frac{\text{Cash} + \text{Marketable securities}}{\text{Current liabilities}}$$

Cash = cash and bank balances; current investments may replace marketable securities (SM 3.7).

∑ **Formula**

Basic defense interval (interval measure)

$$\text{BDI} = \frac{C + R + M}{\text{Daily operating expenses}} = \frac{\text{CA} - \text{Prepaid} - \text{Inventories}}{\text{Daily operating expenses}}$$

C = cash and bank balances, R = net receivables, M = marketable securities; CA = current assets; prepaid = prepaid expenses (SM 3.7).

$$\text{Daily operating expenses} = \frac{\text{COGS} + S - N}{\text{No. of days in a year}}$$

S = selling, administration and other general expenses; N = depreciation and other non-cash expenditure. The answer is in days; take 360 days unless the question says otherwise.

∑ **Formula**

Net working capital

$$\text{NWC} = \text{CA} - \text{CL}$$

NWC = net working capital; CA = current assets; CL = current liabilities. The Study Material's definition adds " (excluding short-term bank borrowing)" to current liabilities (SM 3.8); see Under the hood.

∑ **Formula**

Equity ratio and debt ratio

$$\text{Equity ratio} = \frac{\text{Shareholders' equity}}{\text{Net assets}}$$

$$\text{Debt ratio} = \frac{\text{Total debt}}{\text{Net assets}}$$

Shareholders' equity = equity share capital + reserves and surplus (excluding fictitious assets). Net assets (capital employed) = net fixed assets + net current assets. Total debt = short and long-term borrowings from financial institutions, debentures and bonds, deferred payment arrangements for capital equipment, bank borrowings, public deposits and any other interest-bearing loan (SM 3.9–3.10).

∑ **Formula**

Debt to equity ratio

$$\text{D/E} = \frac{\text{Total outside liabilities}}{\text{Shareholders' equity}} \quad \text{or} \quad \frac{\text{Long-term debt}}{\text{Shareholders' equity}}$$

The first counts both current and non-current liabilities; sometimes only interest-bearing long-term debt is used (SM 3.10).

∑ **Formula**

Debt to total assets, capital gearing and proprietary ratio

$$\text{Debt to total assets} = \frac{\text{Total outside liabilities}}{\text{Total assets}}$$

$$\text{Capital gearing} = \frac{P + D + B}{E + R - L}$$

$$\text{Proprietary ratio} = \frac{\text{Proprietary fund}}{\text{Total assets}}$$

Debt to total assets may use total debt in the numerator. For gearing, P = preference share capital, D = debentures, B = other borrowed funds, E = equity share capital, R = reserves and surplus, L = losses. Proprietary fund = equity share capital + preference share capital + reserves and surplus; total assets exclude fictitious assets and losses (SM 3.10–3.11).

∑ **Formula**

Coverage ratios

$$\text{DSCR} = \frac{\text{Earnings for debt service}}{\text{Interest} + \text{Instalments}}$$

$$\text{Interest coverage} = \frac{\text{EBIT}}{\text{Interest}}$$

Earnings (available) for debt service = EAT + depreciation and other non-cash operating expenses +

interest + other adjustments such as loss on sale of fixed assets (SM 3.11–3.12).

$$\text{Preference dividend coverage} = \frac{\text{EAT}}{\text{Preference dividend}}$$

$$\text{Equity dividend coverage} = \frac{\text{EAT} - \text{Preference dividend}}{\text{Equity dividend}}$$

$$\text{Fixed charges coverage} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest} + \text{Repayment of loan}}$$

ICAI's norms (SM 3.39): DSCR 2; interest, preference dividend and fixed charges coverage above 1.

Note

ICAI's notes for calculating ratios (SM 3.13)

1. EBIT = PBIT (profit before interest and taxes); EAT = PAT (profit after taxes); EBT = PBT (profit before taxes).
2. Ratios shall be calculated based on requirement and availability of information and may deviate from the original formulae. If required, assumptions should be given.
3. The numerator should be taken in correspondence with the denominator, and vice versa.

"Discuss liquidity ratios" and "Discuss gearing ratio"

The chapter's own theory questions include "Discuss Stock Turnover ratio and Gearing ratio" and "Discuss the financial ratios for evaluating company performance on ... liquidity position" (SM 3.65). For each ratio write, in this order:

1. What it measures, in ICAI's words (liquidity: ability to pay short-term liabilities; gearing: proportion of fixed interest or dividend bearing capital to equity shareholders' funds).
2. The formula, with the components defined (what is a quick asset; what is in the denominator of gearing).
3. The interpretation and the norm where ICAI gives one (2:1, 1:1; higher gearing may indicate more risk and signifies the advantage of financial leverage to equity shareholders).
4. Who uses it (short-term creditors and banks for liquidity; long-term lenders for leverage).
5. A one-line caution: a ratio needs comparison (past years, competitors, industry), and liquidity ratios should be read with the turnover ratios.

Three answers to "calculate the DSCR"

Loses marks

"DSCR = 3.08."

A bare figure: no formula, no basis, nothing to give method marks for.

Passes

"DSCR = 37 ÷ 12 = 3.08 times."

Right, but the examiner cannot see that 37 is profit after tax + depreciation + interest.

Full marks

Working note for earnings available for debt service (24 + 9 + 4 = 37) and for debt service (4 + 8 = 12); formula; result in "times"; one line against the norm of 2 ("favourable").

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the prepaid insurance that paid a supplier**
   Fix: deduct inventories **and** prepaid expenses (SM 3.6).

2. **Interest coverage on profit after tax**
   Fix: the numerator is EBIT, because interest is paid before tax and is tax-deductible.

3. **The DSCR that forgot the principal**
   Fix: the denominator is interest + instalments due; "debt service" means both.

4. **The preference shares that became equity**
   Fix: preference capital carries a fixed dividend, so it goes in the **numerator** with the debt; the denominator is equity share capital + reserves − losses.

5. **"Higher current ratio, better company"**
   Fix: say that both too little and too much liquidity are bad (SM 3.5), and check the turnover ratios before praising it.

RECAP

- Current ratio = CA ÷ CL (norm 2:1); quick ratio = (CA − inventories − prepaid expenses) ÷ CL (norm 1:1); cash ratio = (cash and bank + marketable securities) ÷ CL.
- Basic defense interval = quick assets ÷ daily cash operating expenses (COGS + selling, admin and general − depreciation, ÷ days): days of survival without revenue.
- Net working capital = CA − CL. Too little and too much liquidity are both bad.

- Debt-equity = total outside liabilities (or long-term debt) ÷ shareholders' equity: the indicator of financial leverage. Say which debt you use.
- Capital gearing = (preference + debentures + other borrowings) ÷ (equity capital + reserves − losses); proprietary ratio = (equity + preference + reserves) ÷ total assets.
- Interest coverage = EBIT ÷ interest; DSCR = (EAT + non-cash expenses + interest) ÷ (interest + instalments), norm 2; fixed charges coverage = (EBIT + depreciation) ÷ (interest + repayment).
- With a current ratio above 1, equal additions to CA and CL lower it; equal payments raise it.
- CL = working capital ÷ (current ratio − 1). (The bank discounts "sab adjust ho jayega" to zero.)

Interactive version: PassBook app, lesson Liquidity and capital structure ratios

2.2 Activity (turnover) ratios

Key idea

An activity ratio divides a year's sales (or cost of goods sold) by an asset. The higher it is, the more efficiently the asset is used; and 360 divided by a turnover gives the number of days the asset holds each rupee: days in stock, days to collect from debtors, days taken to pay creditors.

FORMATS AND FORMULAS

Reproduce these from memory, with the numerator that belongs to each denominator.

Ratio	Numerator	Denominator
Total assets turnover	Sales or COGS	Total assets
Fixed assets turnover	Sales or COGS	Fixed assets
Capital (net assets) turnover	Sales or COGS	Net assets
Current assets turnover	Sales or COGS	Current assets
Working capital turnover	Sales or COGS	Working capital
Inventory turnover	COGS or sales	Average inventory

Ratio	Numerator	Denominator
Raw material inventory turnover	Raw material consumed	Average raw material stock
Receivables turnover	Credit sales	Average receivables
Payables turnover	Net credit purchases	Average payables

∑ **Formula**

Asset-based turnovers

$$\text{Turnover} = \frac{\text{Sales or COGS}}{\text{Asset base}}$$

Asset base = total assets, fixed assets, net assets (capital employed), current assets or working capital; their averages may be used (SM 3.14–3.16).

∑ **Formula**

Inventory turnover

$$\text{Inventory turnover} = \frac{\text{COGS}}{\text{Average inventory}}$$

$$\text{Average inventory} = \frac{\text{Opening stock} + \text{Closing stock}}{2}$$

$$\text{Raw material inventory turnover} = \frac{\text{Raw material consumed}}{\text{Average raw material stock}}$$

ICAI prints the numerator as "Cost of Goods Sold / Sales" (SM 3.16): use COGS unless the question says "based on sales".

∑ **Formula**

Receivables turnover and average collection period

$$\text{Receivables turnover} = \frac{\text{Credit sales}}{\text{Average receivables}}$$

$$\text{Average collection period} = \frac{\text{Average receivables}}{\text{Average daily credit sales}} = \frac{360 \text{ days}}{\text{Receivables turnover}}$$

Average daily credit sales = credit sales ÷ number of days in the year (say 360); with months or weeks, use 12 or 52 in place of 360 (SM 3.17).

∑ **Formula**

Payables turnover and average payment period

$$\text{Payables turnover} = \frac{\text{Net credit purchases}}{\text{Average payables}}$$

$$\text{Average payment period} = \frac{\text{Average payables}}{\text{Average daily credit purchases}} = \frac{360 \text{ days}}{\text{Payables turnover}}$$

Net credit purchases are the year's (annual) credit purchases; 12 months or 52 weeks may replace 360 days (SM 3.17).

∑ **Formula**

Purchases from the trading account

$$\text{Purchases} = \text{COGS} + \text{Closing} - \text{Opening stock}$$

Needed when a question gives sales, gross profit and stocks but not purchases.

ⓘ **Note**

ICAI's notes for turnover ratios (SM 3.18)

1. Only selling and distribution expenses differentiate cost of goods sold (COGS) and cost of sales (COS); in their absence, COGS = cost of sales.
2. Cost of goods sold or cost of sales can be used to calculate turnover ratios, eliminating the profit part.
3. Averages of total, fixed, current or net assets or working capital can also be taken in the denominator; when average figures are available they may be preferred.
4. Ratios shall be calculated based on requirement and availability of information and may deviate from the original formulae; if required, assumptions should be given.

ANSWER FRAME

"Discuss stock turnover ratio" and "ratios for

operating efficiency"

The chapter's theory questions include "Discuss Stock Turnover ratio and Gearing ratio" and "Discuss the financial ratios for evaluating company performance on operating efficiency and liquidity position aspects" (SM 3.65). For a turnover ratio, write:

1. What it measures: the efficiency with which the firm manages and utilises the asset (activity ratios are asset management ratios).
2. The formula, with the numerator that matches the denominator (COGS for stock, credit sales for debtors, credit purchases for creditors) and the average balance.
3. The conversion to days or months.
4. The interpretation in ICAI's words (a high stock turnover is good for liquidity; a low one means stock stays on the shelf; a low debtors' turnover reflects liberal credit terms).
5. The comparison that gives it meaning: previous years, competitors, the industry.

For **operating efficiency** (SM 3.34): ratio analysis throws light on the degree of efficiency in the management and utilisation of assets; the activity ratios measure this operational efficiency; and the solvency of a firm is, in the ultimate analysis, dependent upon the sales revenues generated by the use of its assets, total as well as its components.

Three answers to "calculate the average collection period"

Loses marks

"ACP = 250 ÷ 1,380 × 360 = 65 days."

Closing debtors used although the opening balance is given; no average, no working.

Passes

"ACP = 195 ÷ 1,380 × 360 = 50.9 days."

Right, but 195 appears from nowhere.

Full marks

Working note: average debtors = (140 + 250) ÷ 2 = 195; average daily credit sales = 1,380 ÷ 360 = 3.83; ACP = 195 ÷ 3.83 = 50.9 days; one line of comment against the previous year or the credit terms.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: stock at cost, sales at selling price**
   Fix: use cost of goods sold for inventory unless the question says "based on sales".

2. **Total sales in the debtors turnover**
   Fix: credit sales ÷ average receivables; take total sales only when credit sales are not given, and say so.

3. **Cost of goods sold in the creditors turnover**
   Fix: derive purchases (COGS + closing stock – opening stock) and use the credit part of them.

4. **The closing balance that pretended to be an average**
   Fix: average whenever you can; use the closing balance only when nothing else is given, and state it.

RECAP

- Activity ratios (efficiency, performance, turnover; "asset management ratios") = sales or COGS ÷ an asset, in "times".
- Total assets, fixed assets, capital (net assets), current assets and working capital turnover: sales or COGS ÷ that base; averages may be used.

- A very high working capital turnover means more working capital will be needed.
- Inventory turnover = COGS ÷ average inventory; raw material turnover = materials consumed ÷ average raw material stock.
- Debtors turnover = credit sales ÷ average receivables; collection period = 360 ÷ turnover (or average receivables ÷ daily credit sales).
- Creditors turnover = credit purchases ÷ average payables; payment period = 360 ÷ turnover. Purchases = COGS + closing − opening stock.
- Balance = annual flow × days ÷ 360. (A full godown is only impressive if it keeps emptying.)

Interactive version: PassBook app, lesson Activity (turnover) ratios

2.3 Profitability and market ratios, and the DuPont analysis

Key idea

Profitability ratios relate profit to sales (margins), to the money invested (ROA, ROCE, ROE), to each share (EPS, DPS) and to the share's price (P/E, yields, market-to-book). Return on investment = margin × turnover, and return on equity = net profit margin × asset turnover × equity multiplier: two companies with the same ROE can be very different businesses.

FORMATS AND FORMULAS

Reproduce these from memory. Every "× 100" ratio is a percentage; the others are in times or rupees.

∑ **Formula**

Margins on sales

$$\text{Gross profit ratio} = \frac{\text{Gross profit}}{\text{Sales}} \times 100$$

$$\text{Net profit ratio} = \frac{\text{EAT}}{\text{Sales}} \times 100$$

$$\text{Pre-tax profit ratio} = \frac{\text{EBT}}{\text{Sales}} \times 100$$

$$\text{Operating profit ratio} = \frac{\text{EBIT}}{\text{Sales}} \times 100$$

Operating profit = sales − COGS − operating expenses (SM 3.20–3.21).

∑ **Formula**

Expenses ratios

$$\text{COGS ratio} = \frac{\text{COGS}}{\text{Sales}} \times 100$$

$$\text{Operating expenses ratio} = \frac{A + S}{\text{Sales}} \times 100$$

$$\text{Operating ratio} = \frac{\text{COGS} + \text{Operating expenses}}{\text{Sales}} \times 100$$

$$\text{Financial expenses ratio} = \frac{\text{Financial expenses}}{\text{Sales}} \times 100$$

A = administrative expenses; S = selling and distribution overheads. Financial expenses exclude taxes, loss due to theft, goods destroyed by fire etc. (SM 3.21).

∑ **Formula**

Return on investment

$$\text{ROI} = \frac{\text{Return}}{\text{Investment}} \times 100 = \frac{\text{Return}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Investment}}$$

(SM 3.22)

∑ **Formula**

Return on assets

$$\text{ROA} = \frac{\text{PAT}}{\text{Average total assets}} \quad \text{or} \quad \frac{\text{PAT} + \text{Interest}}{\text{Average total assets}}$$

$$\text{ROTA} = \frac{\text{EBIT}(1-t)}{\text{Average total assets}}$$

$$\text{RONA} = \frac{\text{EBIT}(1-t)}{\text{Average net assets}}$$

PAT = net profit after taxes; t = tax rate. Average tangible assets or average fixed assets may replace total assets in the first two; total assets are sometimes used instead of average assets (SM 3.23).

∑ **Formula**

Return on capital employed

$$\text{ROCE (pre-tax)} = \frac{\text{EBIT}}{\text{Capital employed}} \times 100$$

$$\text{ROCE (post-tax)} = \frac{\text{EBIT}(1-t)}{\text{Capital employed}} \times 100$$

Sometimes (PAT + interest) ÷ capital employed. Capital employed = total assets − current liabilities = fixed assets + working capital = equity + long-term debt; include intangibles (goodwill, patents, trade marks), exclude fictitious assets (deferred expenses); use average capital employed if available (SM 3.24, 3.30).

∑ **Formula**

Return on equity and the DuPont model

$$\text{ROE} = \frac{\text{PAT} - \text{Preference dividend}}{\text{Equity shareholders funds}} \times 100$$

$$\text{ROE} = \text{Net profit margin} \times \text{Asset turnover} \times \text{Equity multiplier}$$

Net profit margin = net income ÷ sales; asset (investment, capital) turnover = sales ÷ investment, assets or capital; equity multiplier = investment, assets or capital ÷ shareholders' equity (SM 3.25–3.26).

∑ Formula

Owner's ratios

$$\text{EPS} = \frac{\text{Equity earnings}}{\text{No. of equity shares}}$$

$$\text{DPS} = \frac{\text{Equity dividend}}{\text{No. of equity shares}}$$

$$\text{Dividend pay-out ratio} = \frac{\text{DPS}}{\text{EPS}}$$

Equity earnings = net profit available to equity shareholders; equity dividend = total dividend paid to equity shareholders; both are divided by the number of equity shares outstanding. In the absence of preference dividend, PAT is the earnings available to equity shareholders (SM 3.27–3.28, 3.30).

∑ Formula

Market ratios

$$\text{P/E ratio} = \frac{\text{MPS}}{\text{EPS}}$$

$$\text{Earnings yield} = \frac{\text{EPS}}{\text{MPS}} \times 100$$

$$\text{Dividend yield} = \frac{\text{DPS}}{\text{MPS}} \times 100 \quad \text{or} \quad \frac{\text{Dividend} \pm \text{Price change}}{\text{Initial share price}} \times 100$$

$$\text{MV/BV} = \frac{\text{Share price}}{\text{Book value per share}} \qquad \text{Q ratio}$$
$$= \frac{\text{Market value of a company}}{\text{Replacement cost of assets}}$$

Share price = the average or the closing share price; book value per share = net worth ÷ number of equity shares; price change = the change in share price over the period. The Q ratio is also written as market value of equity and liabilities ÷ estimated replacement cost of assets (SM 3.28–3.29).

ANSWER FRAME

"Discuss the DuPont model" (SM 3.65, theory questions 4 and 6)

1. Origin: created in 1919 by a finance executive at E.I. Du Pont de Nemours and Co.; used worldwide as the basis of the components of ROE.
2. The model: ROE = net profit margin × asset (investment) turnover × equity multiplier.
3. Each component, its formula and meaning: net profit margin (profit for each rupee of revenue; a safety cushion); asset turnover (how effectively assets are converted into sales; inversely related to margin); equity multiplier (financial leverage; shows what portion of ROE results from debt).
4. Use: by examining each input, the sources of a company's ROE can be discovered and compared with

competitors; a high ROE from leverage is less attractive than one from margins and turnover.

5. A two-line numerical illustration.

"Discuss any three ratios computed for investment analysis" (SM 3.65, theory question 1)

The Study Material's answer key points to paragraph 3.4.2, the return on investment ratios. For each of ROA, ROCE and ROE: definition, formula with the components named, interpretation (ROCE above the borrowing rate; ROE as an indicator of profitability and potential growth), and who uses it.

Three answers to "calculate the return on equity"

Loses marks

"ROE = 3.75 ÷ 60 = 6.25%."

Total assets in the denominator: that is a return on assets.

Passes

"ROE = 3.75 ÷ 27.50 = 13.64%."

Correct, but the equity figure is unexplained.

Full marks

Working note: equity shareholders' funds = share capital ₹10 crore + reserves ₹17.50 crore = ₹27.50 crore; PAT from the statement of profit; ROE = (PAT − preference dividend) ÷ equity shareholders' funds × 100 = 13.64%; DuPont check if asked.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the preference dividend that went to the equity shareholders**
   Fix: deduct the preference dividend first; only what is left is available to equity shareholders.

2. **The return that did not match its investment**
   Fix: returns before interest (EBIT, EBIT(1 – t), PAT + interest) go over assets or capital employed; returns after interest and preference dividend go over equity shareholders' funds.

3. **DuPont with two engines**
   Fix: that product is a return on assets; ROE needs the equity multiplier too.

4. **Fictitious assets in capital employed**
   Fix: include intangibles such as goodwill, patents and trade marks; exclude fictitious assets (SM 3.24).

RECAP

- Gross profit, operating profit (EBIT) and net profit ratios = that profit ÷ sales × 100; operating ratio = (COGS + operating expenses) ÷ sales, and operating ratio + operating profit ratio = 100%.
- ROI = margin × investment turnover; ROA = PAT (or PAT + interest, or EBIT(1 − t)) ÷ average assets.
- ROCE = EBIT (pre-tax) or EBIT(1 − t) (post-tax) ÷ capital employed; it should exceed the borrowing rate; intangibles in, fictitious assets out.
- ROE = (PAT − preference dividend) ÷ equity shareholders' funds.
- DuPont: ROE = net profit margin × asset turnover × equity multiplier (assets ÷ equity); the multiplier shows the part of ROE that comes from debt.
- EPS = earnings for equity ÷ equity shares; DPS; pay-out = DPS ÷ EPS.
- P/E = MPS ÷ EPS; dividend yield = DPS ÷ MPS (or dividend ± price change ÷ initial price); earnings yield = EPS ÷ MPS; MV/BV; Q ratio = market value ÷ replacement cost (1 = equilibrium).
- Same ROE, different businesses: ask which engine produced it. (Borrowed money sends a bill every month.)

Interactive version: PassBook app, lesson Profitability and market ratios, and the DuPont analysis

2.4 Building financial statements from ratios; limitations of ratios

Key idea

Rebuilding statements from ratios is solving equations: start from a rupee figure, use each ratio to find one new figure, solve two ratios together when they share two unknowns, call sales "x" when no chain starts from the figure you have, and let total assets = total liabilities fix the balancing figure. Then remember that a ratio, however exact, is a clue for an expert, not a conclusion.

FORMATS AND FORMULAS

Reproduce the two formats from memory; ICAI's suggested answers present the rebuilt statements in these layouts, with each balancing figure labelled (bal. fig. = balancing figure).

Trading and Profit and Loss Account for the year ended ...

Dr. Particulars	₹
To Opening stock	xxx
To Purchases (balancing figure)	xxx
To Gross profit c/d	xxx
	xxx
To Operating expenses (balancing figure)	xxx
To Interest	xxx
To Provision for tax	xxx
To Net profit	xxx
	xxx

Cr. Particulars	₹
By Sales	xxx
By Closing stock	xxx
	xxx
By Gross profit b/d	xxx
	xxx

Liabilities	₹	Assets	₹
Share capital	xxx	Fixed assets	xxx
Reserves and surplus	xxx	Current assets:	
Long-term debt	xxx	Stock	xxx
Current liabilities	xxx	Debtors	xxx
		Cash (bal. fig.)	xxx
Total	**xxx**	**Total**	**xxx**

∑ **Formula**

The identities behind every rebuild

$$\text{Total assets} = \text{Total liabilities}$$

$$\text{GP} = \text{Sales} - \text{COGS}$$

$$\text{COGS} = \text{Opening stock} + \text{Purchases} - \text{Closing stock}$$

$$\text{Net worth} = \text{Share capital} + \text{Reserves}$$

$$\text{Average} = \frac{\text{Opening} + \text{Closing}}{2}$$

Net worth here means equity shareholders' funds; total liabilities include capital unless the question says otherwise.

∑ **Formula**

Balances from ratios

$$\text{Debtors} = \frac{\text{Credit sales} \times \text{Days}}{360}$$

$$\text{Stock} = \frac{\text{COGS}}{\text{Stock turnover}}$$

$$\text{CL} = \frac{\text{Working capital}}{\text{Current ratio} - 1}$$

$$\text{Fixed assets} = \frac{\text{Sales}}{\text{Fixed assets turnover}}$$

Days = the collection period in days. Use the basis the question gives (sales or COGS; closing or average; 360 or 365 days).

ANSWER FRAME

"Explain briefly the limitations of financial ratios"

1. One line for each of the eight limitations, in ICAI's order, with its explanation (aggregate data of diverse divisions; historical cost under inflation; seasonal stock and year-end figures; year-end window dressing of popular ratios; non-comparable policies and periods; no standard set of ratios; a high or low ratio not good or bad in itself; inter-related ratios and multivariate analysis).
2. The closing line: financial ratios provide clues but not conclusions; they are tools only in the hands of

experts, as there is no standard ready-made interpretation.

3. For a case ("Comment", "Advise"), apply each relevant limitation to the facts, then conclude what the analyst should do.

Three answers to a "prepare the balance sheet" question

Loses marks

A balance sheet with the right figures and no working notes.

If one figure is wrong, nothing in the chain earns method marks, and the examiner cannot see which ratio was used.

Passes

Working notes for most figures, but the balancing figure is not labelled and the statement is not in the format asked.

The examiner has to search for how cash was found.

Full marks

Numbered working notes in the order of the chain, each with the ratio, the figures and the result; assumptions stated (closing or average, 360 days); statements in the asked format with "(balancing figure)" labelled; two ratios re-checked.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the balance sheet that balanced by accident**
   Fix: find every figure a ratio gives first; only the one figure no ratio gives may balance the sheet, and recompute two ratios at the end.

2. **Gross profit on cost used as gross profit on sales**
   Fix: convert first: 25 ÷ (100 + 25) = 20% of sales, so COGS = 80% of sales.

3. **Closing stock where the ratio uses the average**
   Fix: when an opening stock is given, work with the average, (opening + closing) ÷ 2; when the average is what you find, closing = 2 × average − opening.

4. **The limitations answer that listed but did not apply**
   Fix: name the limitation, quote the fact from the case that shows it, and say what the analyst should do.

RECAP

- Method: sort anchors from ratios; one ratio, one new figure; two ratios for two unknowns; "let sales be x" when the anchor enters only through the totals; balancing figure last; verify two ratios.
- CL = working capital ÷ (current ratio − 1); with inventory I, CL = I ÷ (current ratio − quick ratio).
- Closing stock = 2 × average stock − opening stock; GP g% on cost = g ÷ (100 + g) of sales.
- Present the rebuilt statements in the asked format with every balancing figure labelled.
- ICAI's six uses of ratio analysis: liquidity, long-term solvency, operating efficiency, overall profitability, inter-firm comparison, budgeting.
- Horizontal analysis compares years; vertical analysis expresses one year's items as % of gross sales or total assets.
- Eight limitations (DISWANDI): diversified lines, inflation, seasonal factors, window dressing, accounting policies and period, no standard set, difficulty to generalise, inter-related ratios.

- Ratios provide clues, not conclusions. (Even after the masala chai.)

Interactive version: PassBook app, lesson Building financial statements from ratios; limitations of ratios

CHAPTER 3

Types of Financing

3.1 Long-term sources: equity, preference, debt and retained earnings

Key idea

Long-term needs are financed from long-term sources. Equity is the costliest but carries no obligation to pay or repay, and dilutes control; debt is the cheapest because interest is tax-deductible, but its interest and repayment are obligatory and it raises financial risk; preference capital sits in between, a hybrid; retained earnings dilute nothing and carry almost no risk.

ANSWER FRAME

"Distinguish between preference shares and debentures" (SM 2.12)

Basis	Preference shares	Debentures
Ownership	Preference share capital is a special kind of share	A debenture is a type of loan which can be raised from the public

Basis

Payment of dividend / interest

Preference shares

Priority as regards a fixed dividend and repayment of capital on winding up

Debentures

Carries a fixed percentage of interest

Basis

Nature

Preference shares

A hybrid form of financing, with some characteristics of equity and some of debt

Debentures

An instrument for raising long-term capital with a fixed period of maturity

Add, for full marks, the points the rest of the chapter supports: preference dividend is not tax-deductible while debenture interest is; debenture interest is payable whether or not there are profits.

"Distinguish between equity shares and preference shares" (SM 2.9)

Basis	Equity share	Preference share
Dividend payment	Paid after preference dividend	Preferred over equity dividend
Rate of dividend	Fluctuating	Fixed
Convertibility	Not convertible	Convertible
Voting rights	Full voting rights	Very limited voting rights

"Discuss the advantages of equity shares" and its cousins

Write the heading, then ICAI's points in ICAI's order, one line of explanation each: four for equity (permanent; financial base; no legal obligation to pay dividends; rights issue), five for preference shares, three for debentures. For "disadvantages": three for equity, two for preference shares, four for debentures. A four-mark question wants four points; a two-mark question wants the definition and two points.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the factory on an overdraft**
   Fix: the basic principle matches the period: long-term needs from long-term sources. An overdraft is repayable on demand.

2. **"Retained earnings are free money"**
   Fix: the decision to plough back depends on the return the company generates against the expected cost of equity (SM 2.10).

3. **Callable and puttable swapped**
   Fix: call = the issuer redeems early (at the call price, generally a premium); put = the investor sells it back.

4. **Preference dividend shown as a tax saving**
   Fix: preference dividend, like equity dividend, is not tax-deductible; only debt interest gives a tax shield.

RECAP

- Fund each need from a source of matching period: long-term (more than 5–10 years), medium-term (1–5 years), short-term (up to one year).
- Three classifications: basic sources (external, internal), maturity of payment, ownership and control (owner's fund, borrowed fund).

- Equity: permanent, owners, residual claim, costliest; advantages: permanent, financial base, no legal obligation to pay dividends, rights issue; disadvantages: risky to investors, EPS dilution, control dilution.
- Preference: priority in fixed dividend and capital; hybrid (dividend not tax-deductible, rate fixed); usually cumulative; six types; no EPS dilution, leverage, no takeover risk.
- Retained earnings: ploughed-back profits of equity shareholders; almost no risk, no dilution; retain if the company's return beats the cost of equity.
- Debentures: trust deed, 3–10 years, secured or unsecured, rated; cheapest (interest tax-deductible), no dilution, good in rising prices; but obligatory, covenants, financial risk, lump-sum repayment.
- Callable: issuer may redeem early at the call price; puttable: investor may sell back. Masala: rupee bond sold abroad.
- Bridge finance: short bank loan pending disbursement of a sanctioned term loan; secured by hypothecation, guarantees, promissory notes; costlier; repaid out of the term loan. (No fryers were sold.)

3.2 Venture capital, lease financing, sale and lease back and other sources

Key idea

Venture capital is long-term, mostly equity finance for new, high-risk, growth-oriented firms, with management support. A financial lease passes the risks and rewards of ownership to the lessee and is non-cancellable; an operating lease leaves them with the lessor and is cancellable. Securitisation pools illiquid loans into marketable securities through a special purpose vehicle.

ANSWER FRAME

"Compare financial lease and operating lease" (SM 2.22–2.23)

Point

Risk and reward

Financial lease

Risk and reward incident to ownership pass to the lessee; the lessor remains only the legal owner

Operating lease

The lessee only gets the use of the asset for a certain time; risk incident to ownership belongs wholly to the lessor

Point

Obsolescence

Financial lease

Lessee bears the risk

Operating lease

Lessor bears the risk

Point

Cancellation

Financial lease

Non-cancellable by either party: the lessor wants his principal back with interest

Operating lease

Kept cancellable by the lessor, who can lease the asset to other willing lessees

Point

Repairs, maintenance

Financial lease

The lessor is only a financier and bears none of these costs

Operating lease

Usually the lessor bears repairs, maintenance or operations

Point

Payout

Financial lease

Usually full payout: one lease repays the cost with interest

Operating lease

Usually non-payout: the lessor leases the same asset again and again

"Explain the methods of venture capital financing"

Four headings in ICAI's order (equity financing, conditional loan, income note, participating debenture),

each with its one number: 49% for equity, 2–15% royalty for the conditional loan, 80–87.50% of project cost for IDBI's VCF under the income note, three phases for the participating debenture.

"What is debt securitisation? Explain the process"

Definition (illiquid assets pooled into marketable securities) → the three functions (origination, pooling with transfer to the SPV, securitisation) → without recourse → off-balance sheet funding for the originator. The car-loan example is a good two-line illustration.

"What benefits and risks would you weigh before a sale and leaseback?"

ICAI's suggested answer to MTP September 2025 Series II Q4(c) (2 marks) gives three benefits and two risks, then a conclusion:

- *Benefits*: (i) immediate cash inflow to ease the cash crunch; (ii) continued use of the asset without ownership, avoiding disruption; (iii) off-balance sheet financing improves capital structure ratios.
- *Risks*: (i) loss of ownership may mean higher long-term costs; (ii) lease obligations add to fixed costs and reduce flexibility.
- *Conclusion*: apply it only if the lease terms are favourable and the company lacks better financing options.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the kiln that was "only rented"**
   Fix: ownership stays with the lessor in every lease. Classify by who bears the risks and rewards (here, obsolescence and maintenance), the period against the economic life, cancellability and whether the rentals pay out the full cost.

2. **The venture capitalist who took control**
   Fix: its equity contribution does not exceed 49% of the total equity, so that effective control and ownership remain with the entrepreneur.

3. **Securitisation answered with only a definition**
   Fix: the suggested answers give the origination, pooling and securitisation functions, the SPV as trustee, "without recourse" and off-balance sheet funding: write all of them.

4. **Equal rentals divided by the pre-tax annuity**
   Fix: rentals are taxable, so divide by (1 – tax rate) × PVAF = 0.70 × 3.696 = 2.5872.

RECAP

- Venture capital: equity finance in new, high-risk, growth-oriented small and medium firms, with sales, networking and management support.

- Methods: equity (at most 49% of total equity), conditional loan (royalty 2–15%, no interest), income note (interest and royalty, low rates), participating debenture (nil, low, then high interest).
- Securitisation: illiquid loans pooled into marketable securities through an SPV; origination, pooling, securitisation; without recourse; off-balance sheet funding.
- Operating lease: short, cancellable, lessor bears maintenance and obsolescence, non-payout. Financial lease: most of the economic life, non-cancellable, lessee bears the risks, full payout, a loan in disguise.
- Other leases: sale and lease back (seller becomes lessee), leveraged (lender paid directly by lessee), sales-aid (manufacturer pays the lessor), close-ended (asset back to lessor) and open-ended (lessee may buy).
- Seed capital assistance: IDBI, interest free, 1% service charge for five years, moratorium up to five years. Deferred payment guarantee: no moratorium.
- Bonds: deep discount and zero coupon sold at a discount; SPN with detachable warrant (4–7 years); inflation bonds add inflation to interest; junk bonds below investment grade, so high yield.
- Lessor's minimum rental: (cost − PV of depreciation tax shields − PV of salvage) ÷ [(1 − t) × PVAF]. (The kiln stays with Tapri; so does its obsolescence.)

3.3 Short-term sources of finance

Key idea

Short-term needs are met from short-term sources. Trade credit (without any explicit cost), accrued expenses and deferred income arise spontaneously, and customer advances are cost free; the rest (commercial paper, bank advances, export credit, inter-corporate and public deposits) must be arranged and carry interest.

ANSWER FRAME

"Show the difference between trade credit and bank overdraft" (RTP September 2025 Q10(b))

Point

Meaning

Trade credit

Credit extended by suppliers, allowing the firm to delay payment for goods or services

Bank overdraft

The customer may withdraw in excess of the credit balance in the current account, up to a limit

Point

Period

Trade credit

Usually 15 to 90 days

Bank overdraft

Repayable on demand, but generally continues for long periods by annual renewal

Point

Cost

Trade credit

A type of interest-free loan, so failing to avail it has an interest cost

Bank overdraft

Interest charged on daily balances

Point

How it arises

Trade credit

Automatically, in the course of business

Bank overdraft

Applied for; operated like cash credit and current accounts

Point

Growth

Trade credit

Grows automatically with the volume of business

Bank overdraft

A fixed limit, within which the borrower draws

"Describe the types of packing credit" and "Explain the forms of post-shipment finance"

The five types in ICAI's order (clean; against hypothecation; against pledge; ECGC guarantee; forward exchange contract), each with its distinguishing line; then the 180-day rule. For post-shipment, the four forms (purchase/discounting of documentary export bills; ECGC guarantee; advance against bills for collection; advance against duty drawbacks and cash subsidy), then the 180-days-from-shipment rule for the last. A 1 + 3

question like May 2026 Q4(b) wants the identification first, in one line.

"Explain in brief the features of commercial paper"

Unsecured promissory note → RBI scheme of 1989 for highly rated corporate borrowers (and primary dealers, all-India financial institutions) → ₹5 lakh or multiples → interest linked to the one-year government bond yield → compulsory credit rating (CRISIL, ICRA, CARE, FITCH Ratings India or another agency specified by the RBI).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: shares for the Diwali stock**
   Fix: short-term needs from short-term sources; for permanent, hard-core working capital the long-term sources are right, but for the seasonal peak they leave idle, costly money after Bhai Dooj.

2. **Clean packing credit "secured on the goods"**
   Fix: clean packing credit has no charge or control over raw material or finished goods; hypothecation and pledge are the second and third types.

3. **The overdraft and the cash credit treated as one**
   Fix: an overdraft is a limit on a current account with interest on daily balances; a cash credit is a limit sanctioned against tradable goods by pledge or hypothecation, interest on the amount actually availed. Both are repayable on demand.

4. **Trade credit called "free"**
   Fix: "without any explicit cost" (SM 2.24), but with an implicit cost passed on in the price or the lost cash discount.

RECAP

- Ten short-term sources (SM order): trade credit, accrued expenses and deferred income, advances from customers, commercial paper, treasury bills, certificates of deposit, bank advances, export finance, inter-corporate deposits and public deposits.
- Trade credit: 15–90 days, open account or bills payable, automatic, no explicit cost, grows with business. Accruals and deferred income: spontaneous. Customer advances: cost free.
- CP: unsecured promissory note, RBI 1989, highly rated issuers, ₹5 lakh multiples, linked to the one-year government bond yield, rated. T-bills: Government of India, 14–364 days. CD: bank, 7 days to one year, saleable in the secondary market.
- Bank advances: loan (all at once), overdraft (daily balances), clean overdraft (personal security), cash

credit (interest on amount availed, against goods), advances against goods, bills purchased/discounted (bills held as security).

- Packing credit: firm order or irrevocable L/C; liquidated within 180 days of commencement; five types (clean, hypothecation, pledge, ECGC guarantee, forward contract).
- Post-shipment: bills purchased/discounted, ECGC guarantee, advance against bills for collection, advance against duty drawback (within 180 days of shipment).
- ICDs: say 6 months, from companies with surplus liquidity. Public deposits: up to 35% of paid-up capital and reserves, 6 months to 3 years, unsecured, not for fixed assets. (The stock is gone by Bhai Dooj; so is the cash credit.)

3.4 Contemporary and international sources of finance

Key idea

International financing reaches foreign capital three ways: loans (foreign currency loans, ECBs), bonds (Euro bonds, foreign bonds, ESG bonds and their variants) and equity (ADRs, GDRs; IDRs the other way round). Contemporary sources such as crowdfunding raise small sums from many people through online platforms, in exchange for equity, for interest, or for nothing.

ANSWER FRAME

"Distinguish between ADRs and GDRs" (the old-scheme May 2022 suggested answer)

Basis	Meaning
ADR	A negotiable instrument issued by a US bank, representing a non-US company's stock traded on a US stock exchange
GDR	A negotiable instrument issued by an international depository bank, representing a foreign company's stock traded worldwide

Basis	Issued in
ADR	The US domestic capital market
GDR	The European capital market

Basis	Listed in
ADR	The American stock exchange
GDR	Non-US stock exchanges

Basis	Relevance
ADR	Foreign companies trade in the US stock market
GDR	Foreign companies trade in any country's stock market other than the US

"Discuss the features of international funding and angel financing"

Two headings. International funding: liberalisation and globalisation; FDI and FII as the two major routes, besides ADRs and GDRs; procurement adapted to foreign investors' requirements. Angel financing: equity financing by a wealthy individual for start-up or expansion in exchange for equity; seeks a higher return than traditional investments; typically 25 to 60 per cent; sometimes the last option for start-ups that do not qualify for bank funding and are too small for venture capital.

"Explain the features of crowd funding"

Definition (from a group of people, typically via the internet) → three forms (equity, P2P loans, donation) → helps substantiate demand before production → three parties (fund raiser, mediator, fund investor) → platform fees.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the Euro bond issued in euros**
   Fix: a Euro bond is not denominated in the currency of the country where it is issued (the term has nothing to do with the euro); the British firm's dollar bond in the USA is a foreign bond (a Yankee bond).

2. **IDRs for Indian companies going abroad**
   Fix: Indian companies go abroad with ADRs and GDRs; IDRs are issued in India by foreign companies.

3. **Crowdfunding and P2P lending as rivals**
   Fix: P2P lending is a category of crowdfunding (the loan form); crowdfunding also covers equity funding and donations.

4. **The green bond that funds anything**
   Fix: green and social bonds finance specific projects; it is the sustainability-linked bond whose proceeds go to general corporate purposes, tied to KPIs.

RECAP

- External finance abroad: commercial banks (FC loans), development banks, discounting of trade bills, international agencies (IFC, IBRD, ADB, IMF), international capital markets (Euro-currency market, export credit, bond issues, financial institutions).
- ECB: commercial loans from non-resident lenders, minimum average maturity 3 years, seven recognised sources, automatic route (no approval) or approval route.
- Euro bond: not in the currency of the country of issue (often bearer), while a foreign bond is a foreign issuer's bond in the local currency (Yankee, Samurai, Bulldog).
- MTN: many lots, one documentation; FRN: up to 7 years; ECP: under one year, usually US dollars; fully hedged bonds: forward sale of all payments.
- ESG bonds: green and social (project-based), sustainability-linked (target-based, KPIs, general purposes).
- Foreign bonds table: FCCB, plain vanilla, convertible FRN, drop lock, variable rate demand obligations,

yield curve note, Euro bond, emerging market bond.

- ADR (US exchange), GDR (Europe, London), IDR (foreign company in India); FDI; Indian companies preferred GDRs because US disclosure is more onerous (US GAAP quarterly accounts).
- Crowdfunding: equity, P2P (unsecured loans with interest), donation; three parties; platforms charge fees. Angel: wealthy individual's equity, 25–60%, when banks will not lend and VC will not look. (Kesar got his photo, in London.)

CHAPTER 4

Cost of Capital

4.1 Cost of debt and of preference capital

Key idea

The cost of debt is the rate at which the net proceeds received today equal the present value of the after-tax interest and the redemption value. Interest is tax-deductible, so Kd is an after-tax cost; preference dividend is not, so Kp gets no tax adjustment.

FORMATS AND FORMULAS

Reproduce each formula, its symbols and the YTM layout from memory.

Formula

Irredeemable debentures (SM 4.7)

$$K_d = \frac{I\,(1-t)}{NP}$$

Kd = cost of debt after tax; I = annual interest payment; NP = net proceeds of debentures or current market price; t = tax rate.

∑ **Formula**

Redeemable debentures: Formula 1, only interest tax-deductible (SM 4.8)

$$K_d = \frac{I(1-t) + \dfrac{RV - NP}{n}}{\dfrac{RV + NP}{2}}$$

RV = redemption value; n = remaining life of the debentures.

∑ **Formula**

Redeemable debentures: Formula 2, discount and premium also tax-deductible (SM 4.9)

$$K_d = \frac{\left(I + \dfrac{RV - NP}{n}\right)(1-t)}{\dfrac{RV + NP}{2}}$$

∑ **Formula**

YTM by interpolation (SM 4.12)

$$K_d = L + \frac{NPV_L}{NPV_L - NPV_H} \times (H - L)$$

L and H = the lower and higher trial rates; NPV_L and NPV_H = the NPVs of the cash flows at those rates.

∑ **Formula**

Zero coupon and deep discount bonds (SM 4.12)

$$RV = NP \times (1 + K_d)^n$$

No interest is paid, so the whole return is the gap between issue price and redemption value.

∑ **Formula**

Value of an amortised bond (SM 4.13)

$$V_B = \sum_{t=1}^{n} \frac{C_t}{(1 + K_d)^t}$$

C_t = principal repaid in year t + interest on the amount outstanding; the cost of such a bond is the rate that equates these uneven flows with its net proceeds.

∑ **Formula**

Convertible debentures: redemption value (SM 4.15)

$$RV = \text{higher of}$$

$$\left(\text{cash value},\ N \times P_0(1 + g)^n\right)$$

N = shares per debenture; P_0 = current share price; g = its expected growth rate; n = years to maturity. Then apply Formula 1 or the YTM.

Formula

Preference shares (SM 4.17–4.18)

$$K_p = \frac{PD}{P_0} \qquad K_p = \frac{PD + \dfrac{RV - NP}{n}}{\dfrac{RV + NP}{2}}$$

PD = annual preference dividend; P_0 or NP = net proceeds from the issue. The first is for irredeemable shares, the second for redeemable ones.

Definition

Flotation cost

The new issue of a security (debt or equity) involves some expenditure in the form of underwriting or brokerage fees, legal and administrative charges, registration fees, printing expenses etc. The sum of all these costs is known as floatation cost. (SM 4.7)

Definition

Net proceeds

Issue price less issue expenses or floatation cost. If the issue price is not given, it may be assumed equal to the current market price; if issue expenses are not given, they are taken as zero. (SM 4.7)

Definition

Yield to maturity (YTM)

The annual return of an investment from the current date till maturity date: the internal rate of return at which the current price of a debt equals the present value of all cash flows. (SM 4.11)

The YTM working, ICAI's layout

Year	Cash flow	PV at L (₹)	PV at H (₹)
0	NP (new issue) or P0 (existing debt)	(NP) × 1.000	(NP) × 1.000
1 to n	Interest net of tax, I(1 – t)	× PVAF (L, n)	× PVAF (H, n)
n	Redemption value, RV	× PVF (L, n)	× PVF (H, n)
	NPV	**+ (NPV at L)**	**– (NPV at H)**

ICAI prints each discount factor in its own column (six columns in all); write it that way on paper. Then interpolate.

Interactive version: PassBook app, lesson Cost of debt and of preference capital

ANSWER FRAME

EXPLAIN the YTM approach of calculating cost of debt (SM 4.11–4.12), in this order:

1. Kd is found by discounting the relevant cash flows using the internal rate of return; YTM is the annual return from the current date till maturity, the IRR at which the current price equals the present value of all cash flows.
2. Cash flows: year 0, net proceeds (new issue) or current market price (existing debt); years 1 to n, interest net of tax, I(1 − t); year n, redemption value.
3. Steps: identify the cash flows; calculate NPVs at two discount rates; calculate the IRR by interpolation.
4. It is superior to the approximation method and preferred in finance; the wider the gap between the two rates, the lower the accuracy.

DISCUSS amortisation of bond (SM 4.13): principal repaid every year rather than at maturity; interest on the outstanding amount, so the cash flows are uneven; value = the sum of each year's interest plus principal discounted at Kd; a two-line example.

EXPLAIN any four features of debentures or bonds (4 marks in MTP Series II for May 2025; SM 4.5–4.6). One heading and two lines each, in ICAI's order:

1. **Face value**: the denominated value of the debenture; interest is calculated on it (9% debentures of ₹100: interest @ 9% on ₹100).

2. **Interest (coupon) rate**: a fixed rate on the face value (except zero coupon and deep discount bonds), paid periodically (annually, semi-annually, etc.).
3. **Maturity period**: a fixed period for redemption; for irredeemable debentures it is not defined and is taken as infinite.
4. **Redemption value**: determined by the debt covenants and paid on the maturity date; it may vary from the face value.
5. **Benefit of tax shield**: interest is allowed as an expense for corporate tax, so it saves tax; the saving is the tax shield.

Write four of the five; ICAI's answer gave all five.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the preference dividend that got a tax refund**

   Fix: preference dividend is an appropriation of after-tax profit and saves no tax; only interest gets the (1 – t).

2. **The face value in the denominator**

   Fix: divide by the net proceeds, or the current market price for existing debt: 6.75 ÷ 90 = 7.50%.

3. **The convertible that forgot to grow**
   Fix: grow the price to maturity, ₹24 × $(1.06)^6$, multiply by the shares per debenture, then take the higher of that and the cash value.

4. **Interest on the wrong base, life from the wrong date**
   Interest is the coupon on the **face value**, not on the issue price or NP; n is the **remaining** life on the date of costing, not the original tenure.

RECAP

- Cost of capital: the return expected by the providers of capital; cost is the market's expectation, not the coupon (SM 4.2–4.3).
- NP = issue price − flotation cost; no issue price, use the market price; existing debt, the market price.
- Irredeemable: $K_d = I(1 - t) \div NP$; $K_p = PD \div NP$.
- Redeemable, approximation: $[I(1 - t) + (RV - NP) \div n] \div [(RV + NP) \div 2]$; if the discount and premium are tax-deductible, $[I + (RV - NP) \div n](1 - t) \div$ average.
- YTM: year 0 NP, years 1 to n $I(1 - t)$, year n RV; NPVs at two rates; interpolate. Superior to the approximation.
- Convertible: RV = higher of cash and shares × $P_0(1 + g)^n$.
- No tax shield on a preference dividend, or in a tax holiday. (The taxman only chips in on interest.)

Interactive version: PassBook app, lesson Cost of debt and of preference capital

4.2 Cost of equity and of retained earnings

Key idea

Equity has a cost even though no payment is fixed: it is the return shareholders expect, read off the share price (dividend or earnings approaches, Gordon's model, realised yield) or built from risk (CAPM). Retained earnings cost the shareholders' forgone return, normally a little below Ke because there is no flotation cost.

FORMATS AND FORMULAS

Reproduce each formula and what its symbols mean from memory.

Formula

Dividend price approach (SM 4.20)

$$K_e = \frac{D}{P_0}$$

D = expected dividend (also written D1); P_0 = market price of equity (ex-dividend).

∑ **Formula**

Earnings price approach (SM 4.21)

$$K_e = \frac{E}{P}$$

E = current earnings per share; P = market price per share.

∑ **Formula**

Growth approach, Gordon's model (SM 4.21)

$$K_e = \frac{D_1}{P_0} + g \qquad K_e = \frac{D_1}{P_0 - F} + g$$

$D_1 = D_0(1+g)$, the next expected dividend; P_0 = current market price per share; g = constant growth rate of dividend; F = flotation cost per share (second form, for newly issued shares).

∑ **Formula**

Growth rate (SM 4.22–4.23)

$$g = \sqrt[n]{\frac{D_0}{D_n}} - 1 \qquad g = b \times r$$

D_0 = current dividend; D_n = dividend n years ago; b = earnings retention rate; r = rate of return on funds invested.

∑ **Formula**

Realised yield, year-by-year data (SM 4.25)

$$1 + Y_t = \frac{D_t + P_t}{P_{t-1}}$$

$$K_e = \left[(1+Y_1)(1+Y_2)\cdots(1+Y_n)\right]^{1/n} - 1$$

Each year's yield is dividend plus closing price over opening price; Ke is their geometric mean. With a purchase price, a series of dividends and a sale price instead, Ke is the IRR of those flows.

∑ **Formula**

CAPM (SM 4.26–4.27)

$$K_e = R_f + \beta\,(R_m - R_f)$$

R_f = risk-free rate of return; β = beta coefficient; R_m = rate of return on market portfolio; $(R_m - R_f)$ = market risk premium.

∑ **Formula**

Cost of retained earnings (SM 4.29)

$$K_r = \frac{D}{P} \qquad K_r = \frac{EPS}{P} \qquad K_r = \frac{D_1}{P_0} + g$$

P = current market price (sometimes the issue price); no flotation cost.

Definition

Systematic and unsystematic risk

Unsystematic risk: company-specific risk related to the company's performance, which can be reduced or eliminated by diversification (diversifiable risk). Systematic risk: the macro-economic or market-specific risk under which a company operates, which cannot be eliminated by diversification (non-diversifiable risk). (SM 4.26)

Definition

Cost of retained earnings

The opportunity cost of dividends foregone by shareholders: the expected rate of return shareholders could have obtained by investing in financial assets. (SM 4.28)

ANSWER FRAME

DISCUSS the dividend price approach and the earnings price approach (2 marks; SM 4.20–4.21):

1. Dividend price approach (dividend valuation model): assumes the dividend per share remains constant forever; cost of equity = expected dividend ÷ market price per share (ex-dividend); Ke = D1 ÷ P0, symbols defined.
2. Earnings price approach: co-relates the earnings of the company with the market price of its share; each

investor expects a certain earning, distributed or not; Ke = E ÷ P (EPS ÷ market price); assumes EPS constant forever.

3. The difference: the earnings price approach seeks to nullify the effect of changes in the dividend policy.

EXPLAIN any four methods of computing the cost of equity: RTP January 2025's answer opens with one line, "the rate of return which equates the present value of expected dividends with the market share price", then gives a heading, one or two lines and the formula for each method. Choose four of the dividend price, earnings price, growth (Gordon) and realised yield approaches and CAPM, and add when each is used (SM 4.19).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: last year's dividend in Gordon's numerator**
   Fix: D1 = D0 × (1 + g) = ₹12.96, and Ke = 13.40%.

2. **Flotation cost charged to retained earnings**
   Fix: retained earnings involve no issue, so no flotation: use the current market price. Flotation belongs only to the new-issue Ke.

3. **The cum-dividend price used as P0**
   Fix: P0 is ex-dividend: subtract the dividend payable from the cum-dividend price first.

4. **Beta times the market return**
   Fix: beta multiplies the premium, Rm − Rf: 7% + 0.9 × 6% = 12.40%. Read the question: "market risk premium" or "equity risk premium" is already Rm − Rf; "market return" is Rm.

RECAP

- Ke = what equity shareholders expect; found as a balancing figure from the price, or from risk (CAPM).
- Dividend price: D ÷ P0; earnings price: EPS ÷ P (nullifies dividend policy); both assume constancy forever.

- Gordon: Ke = D1 ÷ P0 + g; new issue D1 ÷ (P0 − F) + g; D1 = D0(1 + g).
- g = $\sqrt[n]{(D0 \div Dn)} - 1$ (or the FVIF table), or g = b × r.
- Realised yield: the IRR of purchase price, dividends and sale price, or the geometric mean of yearly yields.
- CAPM: Ke = Rf + β(Rm − Rf); beta prices systematic risk only; usable even when the entity is suffering losses.
- Kr: same formulas at the market price, no flotation; normally Kr < Ke. (Free money has the longest invoice.)

Interactive version: PassBook app, lesson Cost of equity and of retained earnings

4.3 WACC and the marginal cost of capital

Key idea

WACC is the after-tax cost of each source weighted by its proportion of total capital, and it is the minimum return at which the company creates value. Market value weights reflect today's prices; marginal weights are the proportions of new funds, and the marginal cost rises at each break point where a cheaper source runs out.

FORMATS AND FORMULAS

Reproduce this layout and the formulas from memory; ICAI's Suggested Answers set out every WACC this way.

Source of capital	Amount (₹)	Weight (a)	After-tax cost (b)	(a) × (b)
Equity share capital	xxx	0.xxxx	Ke	xxx
Retained earnings	xxx	0.xxxx	Kr	xxx
Preference share capital	xxx	0.xxxx	Kp	xxx
Debentures / long-term debt	xxx	0.xxxx	Kd	xxx
Total	**xxx**	**1.0000**		**WACC**

Book value weights take the balance sheet amounts. Market value weights take number × market price for each security, with the market value of equity split between share capital and retained earnings in their book value ratio. Marginal weights take the proportions of the new funds.

∑ **Formula**

WACC

$$K_o = W_e K_e + W_r K_r + W_p K_p + W_d K_d$$

K_o = WACC (overall cost of capital); each W = that source's amount ÷ total capital, so the weights add up to 1; each K = that source's after-tax cost.

Σ **Formula**

Market value of equity split (SM 4.33)

$$MV_{EC} = MV_E \times \frac{BV_{EC}}{BV_{EC} + BV_{RE}}$$

MV_E = market value of equity shares (number × price); BV_{EC}, BV_{RE} = book values of paid-up equity capital and retained earnings. The rest of MV_E is the market value of retained earnings.

Σ **Formula**

Marginal cost of capital and break point (SM 4.38; Illustration 18, SM 4.39–4.41)

$$MCC = \sum(\text{proportion of new funds} \times \text{marginal cost of the source})$$

$$\text{Break point} = \frac{\text{cheaper source available}}{\text{its proportion}}$$

Below the first break point, retained earnings and the cheapest debt are used; above it, the dearer component cost replaces the cheaper one.

Definition

Weighted average cost of capital

The weighted average after-tax costs of the individual components of the firm's capital structure: the after-tax cost of each debt and equity is calculated separately and added together to a single overall cost of capital. (SM 4.31)

Definition

Marginal cost of capital

The cost of raising an additional rupee of capital; since capital is raised in substantial amount in practice, it is the cost incurred in raising new funds. It is derived when the average cost of capital is calculated using the marginal weights, which represent the proportion of funds the firm intends to employ. (SM 4.38)

ANSWER FRAME

DISCUSS the meaning of weighted average cost of capital. ILLUSTRATE with an example (SM 4.31–4.32):

1. A company mixes sources to balance financial risk, ownership control and cost, so the cost of total capital is the weighted average of the costs of the individual sources.
2. Definition: the weighted average after-tax costs of the individual components of the firm's capital structure; also called the overall cost of capital; it depends on the capital structure.
3. The four steps (total capital; proportions; proportion × cost; add).
4. A four-line example table (ICAI's: retained earnings 25% × 10%, equity 10% × 11%, preference 15% × 9%, debt 50% × 6% = 7.95%).
5. Use: the discount rate for DCF and NPV, the hurdle rate; returns above WACC create value.

What is the DIFFERENCE between book value and market value weights? The five-point table above, in ICAI's order, with one line on retained earnings (included in book value of equity; under market values, apportioned out of the market value of equity).

DISCUSS marginal cost of capital (SM 4.38): definition; marginal weights = proportions of new funds, so no BV/MV choice; equal to the average cost if proportions and component costs are unchanged; component costs rise beyond certain levels (debt example, external equity with flotation costs); then the average cost rises and the marginal cost rises faster.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the cheapest source as the hurdle rate**
   Fix: the hurdle rate is the WACC of the mix that funds the project, not the cost of the source used this time.

2. **Market value weights with retained earnings at book**
   Fix: split the market value of equity in the book value ratio of equity capital to retained earnings.

3. **Pre-tax interest in the WACC**
   Fix: every cost in a WACC is after tax: Kd = I(1 – t) or its redeemable form; preference and equity costs need no adjustment.

4. **The break point that forgot the proportion**
   Fix: retained earnings fund only the equity part (60%) of each rupee: the break point is ₹72 lakh ÷ 0.60 = ₹120 lakh of total new funds.

RECAP

- Cost of capital: the return expected by providers of capital; also called the cut-off rate, hurdle rate, minimum rate of return; used for investment evaluation, financing decisions and credit policy.

- Five factors: supply and demand, investor preferences, risk and return, inflation, exchange rates.
- WACC = Σ weight × after-tax cost; weights add up to 1; it is the minimum return that creates value.
- BV weights: balance sheet figures, reserves inside equity. MV weights: number × market price; split the market value of equity between share capital and retained earnings in their book value ratio.
- MCC: the cost of an additional rupee, with marginal weights (proportions of new funds); no BV/MV choice.
- Break point = cheaper source available ÷ its proportion; beyond it, the MCC steps up, faster than the average.
- Each tranche keeps its own cost; after new debt, recompute Ke at the new price. (The blend costs more once the Assam runs out.)

Interactive version: PassBook app, lesson WACC and the marginal cost of capital

CHAPTER 5

Financing Decisions: Capital Structure

5.1 Capital structure theories: NI, NOI, traditional and Modigliani-Miller

Key idea

A firm is worth its operating profit capitalised at its overall cost of capital, so the debt-equity mix matters only if it changes that overall cost. NI: it falls as debt rises. NOI and MM without tax: it stays constant, because the cost of equity rises just enough to cancel the cheap debt. Traditional: it first falls, then rises, so there is an optimum. MM with tax: borrowing adds value equal to the tax the interest saves, the debt times the tax rate.

FORMATS AND FORMULAS

Reproduce these two statements from memory: almost every numerical on this topic is one of them.

Value under the NI approach (Ke given)	₹
EBIT	xxx
Less: Interest on debt	(xxx)
Earnings available for equity holders (NI)	xxx
Equity capitalisation rate (Ke)	x%
Market value of equity (S) = NI ÷ Ke	xxx
Market value of debt (D)	xxx
Total value of firm V = S + D	**xxx**
Overall cost of capital Ko = EBIT ÷ V	x%

Value under the NOI approach or MM without tax (Ko given)	₹
EBIT (NOI)	xxx
Overall cost of capital (Ko)	x%
Value of the firm V = EBIT ÷ Ko	**xxx**
Less: Market value of debt (D)	(xxx)
Market value of equity (S)	xxx
Earnings available for equity holders (EBIT − interest)	xxx
Cost of equity Ke = NI ÷ S	x%

∑ **Formula**

Overall cost of capital

$$K_o = K_d \times \frac{D}{D+S} + K_e \times \frac{S}{D+S} \qquad = \frac{\text{EBIT}}{V}$$

Ko is the WACC; D and S are market values of debt and equity (SM 5.3). With tax, use the post-tax Kd in the weighted average.

∑ **Formula**

MM without tax: cost of equity of a levered firm

$$K_e = K_o + (K_o - K_d) \times \frac{\text{Debt}}{\text{Equity}}$$

Ko equals the Ke of the identical unlevered firm (SM 5.16).

∑ **Formula**

MM with tax: value of a levered firm

$$V_g = V_u + \text{TB} \qquad V_u = \frac{\text{EBIT}\,(1-t)}{K_{eu}}$$

TB = present value of tax shields; for permanent debt, debt × t (SM 5.18, 5.69).

Formula

MM with tax: cost of equity and WACC of a levered firm

$$K_{eg} = K_{eu} + (K_{eu} - K_d) \quad \times \frac{\text{Debt}\,(1-t)}{\text{Equity}}$$

$$K_{og} = K_{eu}\,(1 - tL)$$

Kd here is the pre-tax rate of interest; L = debt ÷ (debt + equity) at market value (SM 5.26–5.27).

Arbitrage in three steps (the layout of ICAI's Suggested Answers): 1. Valuation of firms (EBIT, interest, earnings for equity, Ke, S, D, V); 2. Investment and borrowings (sell, borrow or lend, buy); 3. Change in return (new income, less interest, less old income = incremental income due to arbitrage).

Definition

Arbitrage

Arbitrage refers to buying asset or security at lower price in one market and selling it at a higher price in another market. As a result, equilibrium is attained in different markets. (SM 5.16–5.17)

Theory questions and the points that earn marks

"Explain the assumptions of the Modigliani-Miller theory" (SM 5.15): perfect capital markets (information free, no transaction costs); rational investors; equivalent risk classes based on business risk; no corporate taxes; plus the general assumptions (two sources, constant total financing, 100% payout, constant business risk, perpetual life).

"Describe the NOI approach and its assumptions" (SM 5.5, 5.11): NOI = EBIT; capital structure irrelevant; Ko independent of leverage; cheaper debt offset by a rising Ke because equity investors seek compensation for greater risk; Ko and Kd constant, Ke rises; value = EBIT ÷ Ko; the seven general assumptions.

"What is the hierarchy of financing under the pecking order theory, and why does it exist?" (SM 5.29; May 2025): internal finance, then debt (secured, unsecured, hybrid), then new equity last; it exists because of asymmetric information, as managers prefer the source that reveals the least information. ICAI's 2025 answer adds: to minimise the cost of capital, keep financial control and avoid negative market signals.

"Explain the trade-off theory" (SM 5.27–5.28): balance benefit of debt (tax) against costs (financial distress: bankruptcy and non-bankruptcy costs; agency costs);

marginal benefit falls and marginal cost rises with debt; optimum capital structure; explains mixed financing.

Three answers to "Calculate the increase in earnings from arbitrage"

Loses marks

"Value of levered firm is higher, so switch. Gain ₹800."

No valuation table, no borrowing: the method marks are gone.

Passes

Valuation table, then sale of ₹90,000 invested in FrostLine.

Forgot the personal loan, so the risk is not the same and the gain comes out wrong.

Full marks

Three headings: valuation of firms; investment and borrowings (borrow 5% of the levered debt "to maintain the level of risk"); change in return. Final line: incremental income due to arbitrage.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the arbitrageur who forgot to borrow**
   Fix: borrow your share of the levered firm's debt (here ₹50,000) and invest it too; the interest on it comes off the new income.

2. **The value of the firm passed off as the value of equity**
   Fix: equity = Vg − debt = ₹1,560 lakh, and Ke = 21.38%.

3. **NI and NOI swapped**
   Fix: read which rate is given: Ke given, start from earnings for equity (NI); Ko given, start from EBIT ÷ Ko (NOI, MM).

4. **Post-tax Kd in the wrong formula**
   Fix: pre-tax Kd inside (Keu − Kd); post-tax Kd in the weighted average.

RECAP

- Value of the firm = EBIT ÷ Ko; Ko = Kd × D/V + Ke × S/V. Capital structure matters only if it moves Ko.
- NI: Ke, Kd constant; S = NI ÷ Ke; V = S + D; Ko falls; use maximum debt.
- NOI: Ko constant; V = EBIT ÷ Ko; S = V − D; Ke = (EBIT − I) ÷ S rises; mix irrelevant.

- Traditional: Kd flat then up, Ke gradual then speedy; Ko falls then rises; optimum at the lowest Ko.
- MM 1958: perfect markets, rational investors, equivalent risk classes, no tax; Ke = Ko + (Ko − Kd) D/E; arbitrage with personal leverage equalises values.
- MM 1963: Vg = Vu + D × t; Ke = Keu + (Keu − Kd) D (1 − t)/E; Ko = Keu (1 − tL).
- Trade-off: tax benefit against distress and agency costs gives an optimum. Pecking order: internal funds, debt, equity last.
- One samosa, however you slice it, unless the taxman only eats from one plate.

Interactive version: PassBook app, lesson Capital structure theories: NI, NOI, traditional and Modigliani-Miller

5.2 EBIT-EPS analysis, indifference point and financial break-even

Key idea

Hold EBIT fixed, deduct each plan's interest, tax and preference dividend, and divide by its number of shares: that is EBIT-EPS analysis. The financial break-even point is the EBIT where EPS is zero; the indifference point is the EBIT where two plans' EPS are equal. Expect EBIT above the indifference point: prefer the plan with more debt. Below it: prefer equity.

FORMATS AND FORMULAS

Reproduce the EPS statement from memory; every EBIT-EPS answer starts with it.

Statement of EPS under alternative financing plans	Plan I (₹)	Plan II (₹)
EBIT	xxx	xxx
Less: Interest (existing + new)	(xxx)	(xxx)
EBT (PBT)	xxx	xxx
Less: Tax	(xxx)	(xxx)
PAT	xxx	xxx
Less: Preference dividend	(xxx)	(xxx)

Statement of EPS under alternative financing plans	Plan I (₹)	Plan II (₹)
Earnings available to equity shareholders	xxx	xxx
Number of equity shares (existing + new)	xxx	xxx
EPS	**xxx**	**xxx**
PE ratio	x	x
MPS = EPS × PE ratio	**xxx**	**xxx**

∑ **Formula**

Earnings per share

$$\text{EPS} = \frac{(\text{EBIT} - I)(1 - t) - \text{PD}}{N}$$

I = interest, t = tax rate, PD = preference dividend, N = number of equity shares (existing + amount raised ÷ issue price).

∑ **Formula**

Financial break-even point

$$\text{Financial BEP} = I + \frac{\text{PD}}{1 - t}$$

If the company has no preference shares, the financial BEP is simply the interest (SM 6.19).

Σ **Formula**

Indifference point between two plans

$$\frac{(\text{EBIT} - I_1)(1-t)}{E_1} = \frac{(\text{EBIT} - I_2)(1-t)}{E_2}$$

E_1, E_2 = number of equity shares and I_1, I_2 = interest under plans 1 and 2 (SM 5.37). With preference shares, deduct PD after tax: $(\text{EBIT} - I_2)(1 - t) - PD_2$ in the numerator.

Σ **Formula**

Market price per share

$$\text{MPS} = \text{EPS} \times \text{PE ratio} = \frac{\text{EPS}}{K_e}$$

(SM 5.32)

Definition

Financial break-even point

Financial break-even point is the minimum level of EBIT needed to satisfy all the fixed financial charges i.e. interests and preference dividends. It denotes the level of EBIT for which the company's EPS equals zero. (SM 5.36)

Definition

Indifference point

The EBIT at which the EPS lines of two alternatives of financing intersect: between the two given alternatives (regardless of leverage in the financial plans), EPS would be the same at that level of EBIT. Also called the EPS equivalency point. (SM 5.37)

Trick

The one-line indifference formula

Write F for each plan's financial BEP (interest + PD ÷ (1 − t)) and E for its number of shares. Then EBIT at indifference = $(E_1 \times F_2 - E_2 \times F_1) \div (E_1 - E_2)$. Tapri, A against C: (2 × 2.9333 − 1.5 × 0) ÷ 0.5 = ₹11.73 crore. Conditions: the same tax rate in both plans and $E_1 \neq E_2$ (equal shares means no point or every point).

ANSWER FRAME

The theory questions

"Discuss financial break-even and EBIT-EPS indifference analysis" (MTP Series I for January 2026; SM 5.36–5.37):

1. Financial BEP: the minimum EBIT needed to satisfy all fixed financial charges (interest and preference dividend); EBIT at which EPS = 0; formula interest + preference dividend ÷ (1 − t).

2. Below it EPS is negative; if expected EBIT is above it, more fixed-cost financing can be taken, otherwise equity is preferred.
3. EBIT-EPS indifference: the EBIT at which EPS is the same under two financing plans; formula $(EBIT - I_1)(1 - t) \div E_1 = (EBIT - I_2)(1 - t) \div E_2$, symbols defined.
4. It is a vital tool for designing the optimal capital structure.

"Explain the relationship between EBIT-EPS-MPS" (MTP Series II for September 2024; SM 5.33): the objective is the highest MPS, which depends on EPS; at a given EBIT, EPS differs with the financing mix because of fixed financial charges; if the return on assets exceeds the cost of fixed-charge funds, EPS rises (favourable leverage, trading on equity), otherwise it falls; debt is preferred to preference shares because of lower cost and tax-deductible interest.

Three answers to "Calculate the indifference point"

Loses marks

"EBIT = ₹27.2 lakh."

No equation, no shares, no interest: nothing to award if a figure slips.

Passes

The equation with figures, but 4,00,000 new shares (₹80 lakh ÷ ₹20) left out the existing 5,00,000.

Method marks only.

Full marks

Working notes for shares (existing + amount ÷ issue price) and interest (existing + new); the equation; the solution; the EPS at the point as a check; a one-line comment on which plan to prefer above and below it.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: new shares at face value**
   Fix: new shares = amount raised ÷ issue price (face value + premium).

2. **Preference dividend treated like interest**
   Fix: it comes out of PAT; in the financial BEP, divide it by (1 – t).

3. **The forgotten old loan**
   Fix: every plan carries the existing interest and the existing shares; only the new money differs.

4. **Highest EPS crowned the winner**
   When the question gives a PE ratio that changes with leverage, the winner is the highest MPS. In Example 2 the all-debt plan has the best EPS and the worst MPS.

RECAP

- EPS = [(EBIT – I)(1 – t) – PD] ÷ N; new shares = amount ÷ issue price.
- Same EBIT for every plan; existing interest and shares go into every plan.
- Financial BEP = I + PD ÷ (1 – t): the EBIT at which EPS = 0.

- Indifference point: $(EBIT - I_1)(1 - t) \div E_1 = (EBIT - I_2)(1 - t) \div E_2$; deduct PD after tax where there are preference shares.
- Above the indifference point: more debt gives more EPS. Below: more equity.
- Same number of shares: no indifference point (different after-tax costs) or every EBIT is one (same costs).
- Debt beats preference shares: its rate is usually lower and its interest is tax-deductible, so it wins even at the same rate.
- Different PE ratios: choose the highest MPS, not the highest EPS. (The pass beats the card only if you ride enough.)

Interactive version: PassBook app, lesson EBIT-EPS analysis, indifference point and financial break-even

5.3 Capital structure planning, determinants and over- and under-capitalisation

Key idea

Capital structure is planned to balance control, risk and cost so as to maximise the value of the firm, guided by seven principles (trading on equity, growth and stability of sales, cost, risk, control, flexibility, other considerations). Over-capitalisation (capital more than earnings justify) and under-capitalisation (capital less than earnings warrant) are both bad; over-capitalisation is the more dangerous.

ANSWER FRAME

"Explain over-capitalisation. What are its causes and consequences?" (4 marks)

Meaning, with the chief sign (falling dividend, interest and share value) → all five causes → all four consequences, each in ICAI's words as listed in section 6 above. In a scenario ("a very high capital base but low EPS"), name the situation in the first line.

"Write a note on under-capitalisation"

Meaning (with the secret reserves case) → three consequences → five effects → three remedies → one line: less dangerous than over-capitalisation, a problem of adjusting capital structure.

Three answers to "Give two consequences of over-capitalisation" (2 marks)

Loses marks

"The company has too much money and will face losses."

No name for the situation, and "losses" is not one of ICAI's consequences.

Passes

"Over-capitalisation. Dividends fall and so does the share price."

Right ideas, loose words.

Full marks

"The situation is over-capitalisation. Consequences: (i) considerable reduction in the rate of dividend and interest payments; (ii) reduction in the market price of shares."

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the bonus issue for an over-capitalised company**
   Fix: over-capitalisation needs reorganisation, buyback, reduction of debenture and creditor claims, or reduction in the value of shares.

2. **The two definitions swapped**
   Fix: over = more capital than needed, earnings insufficient; under = capitalisation lower than the earning capacity warrants.

3. **Under-capitalisation read as "short of cash"**
   Fix: in ICAI's sense it is a company whose earnings are high for its small book capital, often because of secret reserves; its symptoms are a high dividend rate and a high market value.

4. **Only "cost" in a capital structure answer**
   Fix: a full answer balances cost against risk and control (the three factors) and, for "factors affecting", walks through all seven principles, ending in a compromise.

RECAP

- Capital structure: the combination of equity shareholders' fund, preference share capital and long-

term external debts; balance control, risk and cost to maximise value.

- The decision: forms of financing, amounts and proportions in total capitalisation.
- Seven principles in order: trading on equity, growth and stability of sales, cost, risk, control, flexibility, other considerations.
- Optimal structure: highest value, lowest cost of capital; found by EBIT-EPS-MPS, indifference point and financial BEP analysis.
- Over-capitalisation: more capital than needed, earnings insufficient; 5 causes, 4 consequences, 4 remedies; chief sign: falling dividend and interest.
- Under-capitalisation: capitalisation below what earnings warrant (secret reserves); high dividend rate and share value; remedies: split, bonus (most appropriate), higher par value.
- Over-capitalisation is the more dangerous. (The empty wedding hall is the real worry; the busy chai stall only needs bigger books.)

CHAPTER 6

Financing Decisions: Leverages

6.1 Operating, financial and combined leverage

Key idea

Fixed costs are the lever. Fixed operating costs make EBIT move faster than sales (operating leverage, business risk); fixed financial charges make EPS move faster than EBIT (financial leverage, financial risk); the two multiply into combined leverage (total risk). Each degree of leverage is a multiplier on a percentage change, and it magnifies falls exactly as much as rises.

FORMATS AND FORMULAS

Reproduce this statement from memory: ICAI prints it as the "chart showing" the three leverages, with a bracket beside the part of the statement each one covers (SM 6.4). The right-hand column marks where each measure starts and ends.

Profitability statement	₹	Leverage
Sales	xxx	DOL and DCL start
Less: Variable cost	(xxx)	
Contribution	**xxx**	

Profitability statement	₹	Leverage
Less: Fixed cost	(xxx)	
Operating profit / EBIT	**xxx**	DOL ends, DFL starts
Less: Interest	(xxx)	
Earnings before tax (EBT)	**xxx**	
Less: Tax	(xxx)	
Profit after tax (PAT)	xxx	
Less: Preference dividend (if any)	(xxx)	
Net earnings available to equity shareholders	xxx	
Number of equity shares (N)	xxx	
Earnings per share (EPS) = PAT ÷ N	**xxx**	DFL and DCL end

∑ **Formula**

Degree of operating leverage

$$\text{DOL} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{\%\ \Delta\ \text{EBIT}}{\%\ \Delta\ \text{sales}} = \frac{1}{\text{Margin of safety ratio}}$$

Contribution = sales − variable cost; EBIT = contribution − fixed operating cost (SM 6.6, 6.8).

∑ **Formula**

Degree of financial leverage

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBT}} = \frac{\%\ \Delta\ \text{EPS}}{\%\ \Delta\ \text{EBIT}}$$

With preference dividend Dp: DFL = EBIT ÷ [EBIT − interest − Dp ÷ (1 − t)] (SM 6.15).

∑ **Formula**

Degree of combined leverage

$$\text{DCL} = \text{DOL} \times \text{DFL} = \frac{\text{Contribution}}{\text{EBT}} = \frac{\%\ \Delta\ \text{EPS}}{\%\ \Delta\ \text{sales}}$$

(SM 6.19–6.20)

Definition

Business risk and financial risk

Business risk refers to the risk associated with the firm's operations: the uncertainty about the future operating income (EBIT). Financial risk refers to the additional risk placed on the firm's shareholders because of use of debt, i.e. the additional risk a shareholder bears when a company uses debt in addition to equity financing. (SM 6.2)

Definition

Operating leverage

The tendency of operating income (EBIT) to change disproportionately with change in sale volume, caused by operating fixed cost; the employment of an asset with a fixed cost so that enough revenue can be generated to cover all the fixed and variable costs. (SM 6.5)

Definition

Financial leverage

The use of funds with a fixed cost in order to increase earnings per share. (SM 6.14)

DOL, DFL and DCL side by side (SM 6.27)

Point

Shows

DOL

business risk

DFL

financial risk

DCL

total (combined) risk

Point

Depends on

DOL

fixed cost

DFL

interest and preference dividend

DCL

fixed cost, interest and preference dividend

Point

Measures % change in

DOL

EBIT from 1% change in sales

DFL

EPS from 1% change in EBIT

DCL

EPS from 1% change in sales

Point

Example

DOL

DOL 3, output +8%: EBIT +24%

DFL

DFL 2, EBIT +5%: EPS +10%

DCL

DCL 6, sales +8%: EPS +48%

Point

Unique for

DOL

each level of output

DFL

each level of EBIT

DCL

each level of sales

Point

Undefined at

DOL

operating BEP

DFL

financial BEP

DCL

financial BEP

ANSWER FRAME

"Differentiate between business risk and financial risk"

The points of ICAI's answer to MTP Series I for January

2026 Q2(b), with the Study Material's definitions (SM 6.2):

Basis	Meaning
Business risk	risk associated with the firm's operations; unavoidable, because of the environment in which the firm operates
Financial risk	additional risk placed on the firm's shareholders as a result of debt use in financing

Basis	Represented by
Business risk	variability of EBIT, influenced by revenues and expenses (demand for products, variations in prices, proportion of fixed cost in total cost)
Financial risk	the use of debt: companies issuing more debt instruments have higher financial risk than companies financed mostly by equity

Basis	Indicated by
Business risk	operating leverage (SM 6.3)
Financial risk	financial leverage (SM 6.3); measured by ratios such as the financial leverage multiplier and total debt to assets ratio

"Explain financial leverage as a double-edged sword" (2 marks)

When the cost of fixed cost funds is less than the return on investment, financial leverage increases return on

equity and EPS, with a tax saving on interest; when the cost of debt is more than the return, it affects return on equity and EPS unfavourably and the firm can be under financial distress. Hence "double edged sword": ROI > interest favourable, ROI < interest unfavourable, ROI = interest neutral.

Three answers to "Calculate the operating leverage"

Loses marks

"OL = 2.33."

No statement: an error in sales or contribution leaves nothing to mark.

Passes

Income statement, then OL = 10.5 ÷ 4.5 = 2.33.

Correct, but no comment where the question asks to "comment".

Full marks

Working note for sales; income statement to EBIT; formula, figures and result to two decimals; one line on what it indicates (business risk, % change in EBIT for 1% change in sales).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: interest in the fixed cost**
   Fix: operating leverage uses fixed operating cost only; interest belongs below EBIT, in financial leverage.

2. **DOL multiplied when it should be divided**
   Fix: DCL = % change in EPS ÷ % change in sales = 45 ÷ 15 = 3, and DFL = DCL ÷ DOL. A fall that "wipes out EPS" is a 100% change: a 25% fall that wipes it out means DCL = 100 ÷ 25 = 4.

3. **Preference dividend ignored without saying so**
   Fix: use EBIT ÷ [EBT − Dp ÷ (1 − t)] (SM 6.15), or state that you have ignored it, as ICAI's alternative answers do.

4. **A DOL of 0.6**
   Fix: DOL = contribution ÷ EBIT, which is at least 1 above the break-even point.

RECAP

- Business risk: uncertainty about EBIT (operating leverage). Financial risk: extra risk to shareholders from debt (financial leverage).
- DOL = contribution ÷ EBIT = % Δ EBIT ÷ % Δ sales = 1 ÷ MOS; function of fixed cost, contribution margin and sales volume.

- DFL = EBIT ÷ EBT = % Δ EPS ÷ % Δ EBIT; with preference shares, EBIT ÷ [EBT − Dp ÷ (1 − t)].
- DCL = DOL × DFL = contribution ÷ EBT = % Δ EPS ÷ % Δ sales.
- No fixed cost: DOL = 1; at the operating BEP: undefined; below it: negative. Same for DFL around the financial BEP. Never between 0 and 1.
- Financial leverage is a double-edged sword: ROI > Kd favourable, ROI < Kd unfavourable.
- Best combination: low DOL, high DFL. Riskiest: high DOL, high DFL.
- "Wipe out EPS" = a 100% fall: DCL = 100 ÷ % fall in sales. (Gears up, gears down.)

Interactive version: PassBook app, lesson Operating, financial and combined leverage

CHAPTER 7

Investment Decisions

7.1 Capital budgeting and estimating relevant cash flows

Key idea

Only post-tax incremental cash flows count: what changes in cash because the project is accepted. Depreciation counts only through the tax it saves; sunk costs, allocated overheads and financing costs stay out.

FORMATS AND FORMULAS

Reproduce these formats from memory; ICAI's suggested answers follow them line by line.

Initial cash outflow: a fresh proposal or an expansion (SM 7.10)

Particulars	Amount	Amount
Cost of new asset(s)		xxx
Add: Installation/set-up costs	xxx	
Add: Investment in working capital	xxx	xxx
Initial cash outflow		**xxx**

Initial cash outflow: a replacement decision (SM 7.10–7.11)

Particulars	Amount	Amount
Cost of new asset(s)		xxx
Add: Installation/set-up costs	xxx	
Add/(less): Increase (decrease) in net working capital level	xxx	
Less: Net proceeds from sale of old assets	(xxx)	
Add/(less): Tax expense (saving/loss) due to sale of old asset	xxx	xxx
Initial cash outflow		**xxx**

Interim cash flow: a new project (SM 7.11)

Particulars	Amount	Amount
Profit after tax (PAT)		xxx
Add: Non-cash expenses (e.g. depreciation)	xxx	
Add/(less): Net decrease (increase) in working capital	xxx	xxx
Interim net cash flow for the period		**xxx**

Interim cash flow: a replacement decision (SM 7.11–7.12)

Particulars	Amount
Net increase (decrease) in operating revenue	xxx
Add/(less): Net decrease (increase) in operating expenses	xxx
Net changes in income before taxes	xxx
Add/(less): Net decrease (increase) in taxes	xxx
Net change in income after taxes	xxx
Add/(less): Net decrease (increase) in depreciation charges	xxx
Incremental net cash flow for the period	**xxx**

The format is reproduced as the Study Material prints it. In the arithmetic, the change in depreciation is deducted (as an operating expense) before the change in taxes is worked out; the last adjustment then reverses it, because depreciation is not cash: an increase in depreciation is added back, a decrease deducted. The pouch-packer example below shows the lines.

Terminal-year net cash flow (SM 7.12)

Particulars	Amount	Amount
Final salvage value (disposal costs) of asset		xxx
Add: Interim cash flow	xxx	
Add/(less): Tax savings (tax expenses) due to sale or disposal of asset (including depreciation)	xxx	
Add: Release of net working capital	xxx	xxx
Terminal year net cash flow		**xxx**

Statement showing the calculation of cash inflow after tax (CFAT) (SM 7.16)

Particulars	(₹)	(₹)
Sales value		xxx
Less: Variable cost		xxx
Contribution		**xxx**
Less: Fixed cost: (a) fixed cash cost (excluding interest)	xxx	
(b) Depreciation	xxx	xxx
Earning before tax (EBT)		xxx
Less: Tax		xxx
Earning after tax (EAT)		xxx
Add: Depreciation		xxx
Cash inflow after tax (CFAT)		**xxx**

∑ **Formula**

Tax shield on depreciation

$$\text{Tax shield} = \text{Depreciation} \times \text{Tax rate}$$

Σ **Formula**

Profit after tax excluding financing cost

$$\text{PBIT} \times (1-t) \quad = \text{PAT} + \text{Interest} \times (1-t)$$

PBIT = profit before interest and tax, PAT = profit after tax (struck after interest), t = tax rate (SM 7.15).

Σ **Formula**

Terminal year: only asset in the block, or other assets in it

$$\text{STCL (STCG)} = W_0 - S$$

$$\text{Depreciation} = (W_0 - S) \times r$$

W_0 = WDV at the beginning of the year of sale, S = sale value, r = depreciation rate. The first when the block ceases; the second when it continues and S is below W_0 (SM 7.13–7.14).

Definition

Capital budgeting

The process of evaluating and selecting long-term investments that are in line with the goal of investor's wealth maximization. (SM 7.69)

Definition

Sunk cost

An outlay of cash that has already been incurred in the past and cannot be reversed in present. (SM 7.9)

Definition

Opportunity cost

Foregoing of a benefit due to choosing an alternative investment option. (SM 7.9)

ANSWER FRAME

The chapter's own theory questions include "Discuss in detail the capital budgeting process" and "Classify various types of capital investment decisions" (SM 7.74). Write:

- **Purpose**: substantial investment; long time period; irreversibility; complex decisions; a sentence each.
- **Process**: planning, evaluation, selection, implementation, control (the three reports), review; a line each.
- **Types**: by firm's existence (replacement and modernisation, expansion, diversification; cost reduction versus revenue expansion) and by situation (mutually exclusive, accept-reject, contingent), each with an example.
- **Treatment of an item**: the principle (incremental, post-tax), the item's treatment in ICAI's words, a two-

line example.

Loses marks

Bare figures: one slip and no method marks survive.

Passes

The CFAT and NPV tables, but the sunk fee and the interest are silently dropped.

Full marks

Items excluded, with reasons; initial outflow, CFAT and terminal year in ICAI's formats; assumptions stated (timing, tax on losses); a one-line advice.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the feasibility fee Kesar paid twice**
   Fix: the fee is sunk and the overheads only allocated. In the samosa example, including them with the interest turns an NPV of ₹20,84,325 into about −₹3.64 lakh: a good project rejected.

2. **The interest that was charged twice**
   Fix: the WACC already contains the cost of debt; take cash flows before interest.

3. **The working capital that never came home**
   Fix: working capital comes back in full at the end unless the question says otherwise; no depreciation is provided on it.

4. **Depreciation in the year the block died**
   Fix: no depreciation that year; opening WDV less sale value is the short-term capital loss.

RECAP

- Capital budgeting: identify projects, estimate post-tax incremental cash flows, select the proposal that maximises return.
- Critical because of substantial investment, long time period, irreversibility, complex decisions.
- Process: planning, evaluation, selection, implementation, control, review.
- Types: replacement and modernisation (cost reduction), expansion, diversification (revenue expansion); mutually exclusive, accept-reject, contingent.
- In: opportunity cost, working capital (out, then released in full), additional investment, incremental overheads, depreciation's tax shield. Out: sunk cost, allocated overheads, interest and dividends.
- Cash flows: initial outflow, interim CFAT (EAT + depreciation), terminal year (salvage, tax on sale, release of working capital).

- Only asset in the block: no depreciation in the year of sale; STCL or STCG on (opening WDV − sale value). Other assets: depreciation on that difference.
- Cash flows and discount rate both post-tax; financing costs live in the WACC. (Two photographs, one difference.)

7.2 Payback, discounted payback and ARR

Key idea

Payback measures how fast the initial cash outlay is recovered: a guide to liquidity and risk, blind to the time value of money and to every cash flow after payback. Discounted payback cures the first blindness, not the second. ARR covers the whole life but uses accounting profit and ignores the time value of money.

FORMATS AND FORMULAS

Reproduce these from memory.

Formula

Payback period, uniform cash flows

$$\text{Payback period} = \frac{I}{C}$$

I = total initial capital investment; C = annual expected after-tax net cash flow (SM 7.19).

∑ **Formula**

Payback period, non-uniform cash flows

$$\text{Payback} = Y + \frac{U}{C_{Y+1}}$$

Y = full years before the outlay is recovered; U = unrecovered outlay (outlay − cumulative inflows of those Y years); C with subscript Y + 1 = cash inflow of the next year.

∑ **Formula**

Payback reciprocal

$$\text{Payback reciprocal} = \frac{\bar{C}}{I} \times 100$$

$\bar{C}$ = average annual cash inflow; I = initial investment (SM 7.22).

∑ **Formula**

Accounting rate of return

$$\text{ARR} = \frac{\text{Average annual net income}}{\text{Investment}}$$

Expressed as a percentage (× 100). Net income = profit after depreciation (and after tax where tax is given). Investment = initial or average investment, as the question says (SM 7.22).

∑ **Formula**

Average investment

$$\frac{1}{2}(I - S) + S + W$$

I = initial investment, S = salvage value, W = additional working capital (SM 7.24). With no working capital this is (I + S) ÷ 2.

The three versions of ARR (SM 7.23–7.24)

Version	Numerator	Denominator
1. Annual basis	Profit after depreciation of each year	Investment at the beginning of that year; then average the yearly rates
2. Total investment basis	Average annual profit	Investment in the beginning
3. Average investment basis	Average annual profit	Average investment

Discounted payback: the working table (SM 7.43)

Year	Cash flow (₹)	PV factor	PV (₹)	Cumulative PV (₹)
1	xxx	x.xxx	xxx	xxx
2	xxx	x.xxx	xxx	xxx
...				

Discounted payback = years before the cumulative PV reaches the outlay + unrecovered amount ÷ that year's PV.

Definition

Payback period

Time required to recover the initial cash-outflow is called pay-back period. (SM 7.18)

Definition

Discounted payback period

The payback period calculated after discounting the cash flows by a predetermined rate. (SM 7.20, 7.42)

ANSWER FRAME

Relevance of payback reciprocal (May 2024 Q4(c), ICAI's points): reciprocal of the payback is a close approximation of the IRR if the life is at least twice the payback period and the cash inflows are equal; a helpful tool for quick estimation of the rate of return; formula:

average annual cash flow ÷ initial investment, or 1 ÷ payback period.

Concept of discounted payback period (MTP Series II for May 2024 Q4(c)): payback is the time to recover the investment from project cash flows, focused on liquidity, ignoring time value and profitability; discounted payback uses cash flows discounted at the cost of capital; the period in which discounted cash flows equal the initial outflow; shorter is better; it still ignores cash flows after that period.

Short note on cut-off rate (MTP Series I for May 2024 Q2(b), ICAI's points): the minimum rate of return the management wishes to have from any project; usually based on the cost of capital; the management gains only if a project returns more than the cut-off rate; so it can be used as the discount rate or opportunity cost rate. Add the Study Material's example: a proposal yielding less than the cut-off rate is automatically rejected (SM 7.26).

Advantages and limitations of payback or ARR: ICAI's lists above, in ICAI's order, a line each.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the payback that rounded itself up**
   Fix: add the fraction: unrecovered ₹6,40,000 ÷ year 4's ₹20,70,000 = 0.31, so 3.31 years. Divide by that year's cash flow, not by the cumulative total.

2. **ARR on cash instead of profit**
   Fix: ARR uses average annual net income after depreciation (₹8,40,000), giving 22.70%.

3. **The average investment that lost its salvage**
   Fix: ½ (initial − salvage) + salvage + working capital = ₹37,00,000; salvage and working capital stay blocked to the end.

4. **The discount that shortened payback**
   Fix: only future inflows are discounted; the year-0 outlay stays at face value, and the discounted payback is never shorter.

RECAP

- Payback = initial investment ÷ annual cash flow after tax (uniform); otherwise cumulate and add the fraction: unrecovered ÷ that year's cash flow.
- Payback: easy, a guide to risk and liquidity; ignores time value, cash flows after payback, long-term

projects.

- Payback reciprocal = average annual cash inflow ÷ initial investment; approximates IRR only if life ≥ 2 × payback and inflows are equal.
- ARR = average annual net income (after depreciation) ÷ initial or average investment; the same base for every project.
- Average investment = ½ (initial − salvage) + salvage + additional working capital.
- Discounted payback: payback on discounted cash flows; considers time value; still ignores later cash flows; never shorter than payback.
- Accept: payback ≤ maximum acceptable; ARR ≥ minimum acceptable (cut-off). Mutually exclusive: least payback, highest ARR.
- Last year of a payback or discounted payback table: add salvage and working capital released to that year's inflow. (The cousin's loan: how soon back, how soon back in today's rupees, what it earned by the book.)

Interactive version: PassBook app, lesson Payback, discounted payback and ARR

7.3 NPV, profitability index and equivalent annual cost

Key idea

NPV converts every cash flow into today's rupees at the cost of capital and nets them: accept a project when NPV ≥ 0, because a positive NPV is the addition to shareholders' wealth. PI is the same test as a ratio; equivalent annual figures make projects with different lives comparable.

FORMATS AND FORMULAS

Reproduce these from memory.

Formula

Net present value

$$\text{NPV} = \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} - I$$

C_t = cash flow of year t, k = discount rate, n = life of the project, I = investment (SM 7.27–7.28). With outflows later in the life: NPV = PV of cash inflows − PV of cash outflows.

∑ **Formula**

Profitability index

$$\text{PI} = \frac{\text{Sum of discounted cash inflows}}{\text{Discounted cash outflow}}$$

The denominator is the initial cash outlay or the total discounted cash outflow, as the case may be (SM 7.31).

∑ **Formula**

Equivalent annualised NPV and equivalent annual cost

$$\text{EAN} = \frac{\text{NPV}}{\text{PVAF}_{k,n}} \qquad \text{EAC} = \frac{\text{PV of costs}}{\text{PVAF}_{k,n}}$$

PVAF = present value annuity factor at the discount rate k for the project's own life n (SM 7.56). Higher EAN or lower EAC wins.

The NPV statement (SM 7.29)

Year	Net cash flow (₹)	PV factor @ k	Discounted cash flow (₹)
0	(xxx)	1.000	(xxx)
1	xxx	x.xxx	xxx
…			
Net present value			**xxx**

For equal yearly flows, one line: cash flow × PVAF for years 1 to n.

Definition

Net present value

The net present value of a project is the amount, in current value of amount, the investment earns after paying cost of capital in each period. (SM 7.27)

ANSWER FRAME

"Discuss NPV. How is it calculated?" (SM 7.74, theory question 2): definition (the amount, in current value, the investment earns after paying cost of capital in each period); formula; ICAI's five steps; decision rule (accept if NPV ≥ 0; highest NPV among mutually exclusive projects); advantages and limitations, a line each.

"Describe the advantages and disadvantages of profitability index" (SM 7.74, question 5): the formula and its other names; two advantages; three limitations; the conclusion that PI cannot be used indiscriminately.

PI and NPV: same decisions? (RTP January 2025 Q11(c), ICAI's points): in most situations the same accept-reject decision (PI > 1 means NPV > 0); conflict in ranking mutually exclusive projects; a two-project example (larger NPV, smaller PI); reason: NPV is absolute, PI a ratio; PI is the better technique under capital rationing.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the ₹4 lakh profit that was ₹62,300**
   Fix: discount each year's cash flow first. At 12% the inflows are worth ₹10,62,300 and the NPV is ₹62,300.

2. **The eight-year machine that won on paper**
   Fix: with unequal lives, compare equivalent annualised NPVs or NPVs over a common life. Here the two disagree, because K4's replacement costs more, and the common-life chain built on the supplier's price decides: K8, at ₹4,14,150 against ₹2,95,373.

3. **Picking by PI when NPV should decide**
   Fix: between mutually exclusive projects, the highest NPV is selected; PI decides only when NPVs are the same, or when capital is rationed.

4. **The cheaper scooter that was dearer**
   Fix: divide each by the PVAF for its own life; the lower equivalent annual cost wins.

RECAP

- Discount every cash flow to today at the cost of capital (WACC); PV factor = $1 \div (1 + k)^t$.

- NPV = PV of cash inflows − PV of cash outflows; accept if NPV ≥ 0; among mutually exclusive projects, the highest NPV.
- NPV is the addition to shareholders' wealth; its limitations: hard calculations, forecasts of cash flows and rate, an absolute measure.
- PI (desirability factor, present value index) = discounted inflows ÷ discounted outflow; accept if PI ≥ 1; it ranks by ratio and fails with indivisible projects under rationing.
- NPV and PI agree on accept-reject; they can conflict on ranking; with NPVs different, NPV decides.
- Unequal lives: replacement chain (common life) or equivalent annualised criterion, NPV ÷ PVAF for each life; the higher wins.
- Costs only: equivalent annual cost = PV of costs ÷ PVAF; the lower wins.
- Use the PV factors the question gives and say which: a printed annuity factor (4.111 at 12% for 6 years) and the sum of three-decimal yearly factors (4.112) differ in the last digits. (Convert the currencies first: a 2030 rupee is not a 2026 rupee.)

Interactive version: PassBook app, lesson NPV, profitability index and equivalent annual cost

7.4 IRR, MIRR and the NPV-IRR conflict

Key idea

IRR is the discount rate at which NPV is zero. Accept a project when its IRR is at least the cost of capital: for a single project this always agrees with NPV. When mutually exclusive projects differ in size, timing or life, IRR and NPV can rank them differently, and the NPV ranking is the one consistent with wealth maximisation.

FORMATS AND FORMULAS

Reproduce these from memory.

Formula

Internal rate of return

$$\sum_{t=1}^{n} \frac{C_t}{(1+r)^t} - I = 0$$

r is the IRR: the rate at which the present value of the inflows C_t equals the investment I (SM 7.32).

∑ **Formula**

Fake payback factor (uniform inflows)

$$A = \frac{\text{Total initial investment}}{\text{Annual (equal) cash inflow}}$$

Look up A in the PVAF table in the row for the project's life (SM 7.33).

∑ **Formula**

Interpolation

$$\text{IRR} = LR + \frac{\text{NPV at } LR}{\text{NPV at } LR - \text{NPV at } HR} \times (HR - LR)$$

or, with present values,

$$\text{IRR} = LR + \frac{\text{PV at } LR - CI}{\text{PV at } LR - \text{PV at } HR} \times (HR - LR)$$

LR = lower rate, HR = higher rate, CI = capital investment (SM 7.33).

∑ **Formula**

Modified internal rate of return

$$\text{MIRR} = \left(\frac{\text{TV}}{\text{Initial outflow}}\right)^{1/n} - 1$$

TV, the terminal value: each inflow compounded to year n at the cost of capital, $(1 + k)^{n-t}$. Alternatively, find outflow ÷ terminal value in the year-n row of the PV table (SM 7.45).

Definition

Modified internal rate of return

Under this method, all cash flows, apart from the initial investment, are brought to the terminal value using an appropriate discount rate (usually the Cost of Capital). This results in a single stream of cash inflow in the terminal year. The discount rate which equates the present value of the terminal cash inflow to the zero year outflow is called the MIRR. (SM 7.43)

IRR working (one table for each trial rate, as in SM 7.35–7.36)

Year	Cash flow (₹)	PV factor @ r	Present value (₹)
0	(xxx)	1.000	(xxx)
1 … n	xxx	x.xxx	xxx
NPV at r			**xxx**

Then the interpolation line, with LR, HR and both NPVs written out.

MIRR working (SM 7.44)

Year	Cash flow (₹)	Reinvestment factor @ k	Terminal value (₹)
1	xxx	$(1 + k)^{n-1}$	xxx
...			
n	xxx	1.000	xxx
Total terminal value			**xxx**

Decision rules

Technique	Independent project	Mutually exclusive projects
IRR	Accept if IRR ≥ k; reject if IRR < k	Maximum IRR (SM 7.52); on a conflict with NPV, the higher NPV
MIRR	Accept if MIRR > required rate; reject if lower	As IRR

ANSWER FRAME

"Write a short note on IRR" (MTP Series II for May 2025, 2 marks; ICAI's points): the rate at which discounted cash inflows equal discounted cash outflows, the rate that discounts the cash flows to zero; it can be stated as discounted inflows ÷ discounted outflows = 1; found by trial and error, since the rate is unknown while the cash

flows are known; compared with the cut-off rate: accept if more, otherwise reject.

"Identify the limitations of IRR" (old scheme May 2022, 4 marks): the four points of SM 7.42 in ICAI's order (tedious with interspersed outflows and multiple IRRs; peculiar results with different inflow/outflow patterns; reinvestment at the IRR; mutually exclusive projects with different outlays).

"Describe MIRR" (SM theory question 6; MTP Series I for January 2025, 2 marks): what it addresses (multiple IRRs, the reinvestment rate, consistency with NPV); the terminal value method in ICAI's words; the single outflow and single inflow; the decision rule.

"Distinguish between NPV and IRR"

Point	NPV	IRR
Result	Absolute amount (₹)	Percentage
Discount rate	Cost of capital, given	Found; then compared with the cost of capital
Reinvestment assumed at	The discount rate	The project's IRR
Multiple results	One NPV at a given rate	Several IRRs possible when signs change
Mutually exclusive projects	Highest NPV; consistent with wealth maximisation	May conflict with NPV (size, timing, lives)

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the minus sign that sank the IRR**
   Fix: the denominator is NPV at LR − NPV at HR = 2,400 − (−1,600) = 4,000, so IRR = 18% + 1.2% = 19.2%.

2. **Choosing the mutually exclusive project with the higher IRR**
   Fix: with mutually exclusive projects that conflict, choose the higher NPV and give the reason: NPV is consistent with wealth maximisation; IRR ignores scale and assumes reinvestment at the IRR.

3. **Compounding the last year's inflow, or the outlay**
   Fix: an inflow of year t takes $(1 + k)^{n-t}$; check that the year-1 inflow has the largest factor.

4. **Reading the fake payback factor as the answer**
   Fix: discount the actual cash flows at two rates and interpolate.

RECAP

- IRR: the discount rate that equates the PV of the expected cash inflows with the initial cash outflow (NPV = 0).
- Uniform inflows: fake payback = investment ÷ annual inflow, read in the PVAF row for the life, then interpolate. Uneven inflows: trial and error at two rates, then interpolate.
- IRR = LR + NPV at LR ÷ (NPV at LR − NPV at HR) × (HR − LR); the IRR lies between LR and HR.
- Accept if IRR ≥ cut-off rate (WACC), reject if lower; for one project this always agrees with NPV ≥ 0.

- NPV and IRR can conflict for mutually exclusive projects: size disparity, time disparity, unequal lives. Follow NPV (wealth maximisation); for unequal lives, the equivalent annualised NPV.
- Reinvestment: NPV at the discount rate, IRR at the IRR; sign changes can give multiple IRRs.
- MIRR (terminal value method): compound inflows to year n at the cost of capital; MIRR = $(\text{TV} \div \text{outlay})^{1/n} - 1$; accept if MIRR > required rate.
- State your factors: once-rounded annuity factors and summed yearly factors differ in the last digit. (A project's own FD rate is impressive only if you can redeposit at it.)

Interactive version: PassBook app, lesson IRR, MIRR and the NPV-IRR conflict

7.5 Capital rationing

Key idea

Under capital rationing, maximise the total NPV the budget can buy. Divisible projects: rank by NPV per rupee (the profitability index) and fill the budget, the last project in part. Indivisible projects: examine the combinations within the budget and select the one with the highest total NPV, even if it skips a higher-ranked project.

FORMATS AND FORMULAS

Reproduce these from memory.

Formula

NPV per rupee of investment

$$\text{NPV per ₹1} = \frac{\text{NPV}}{\text{Investment}}$$

$$\text{PI} = 1 + \frac{\text{NPV}}{\text{Investment}}$$

The PI form holds when the whole outlay is made at year 0; both give the same ranking (SM 7.53–7.54).

∑ Formula

NPV of a part project

$$\text{NPV of the part taken} = \text{NPV} \times \frac{\text{Funds remaining}}{\text{Investment}}$$

Step 1. Ranking (SM 7.54)

Project	Investment (₹)	NPV (₹)	NPV per ₹1	Rank
A	xxx	xxx	x.xx	n
…				

Step 2 (divisible). Building up the programme

Project (in rank order)	Investment (₹)	NPV (₹)
Rank 1	xxx	xxx
…		
Last: part (fraction)	xxx	xxx
Total	**Budget**	**xxx**

Step 2 (indivisible). Combinations within the limit

Combination	Investment (₹)	Total NPV (₹)
A + B + ...	xxx	xxx
Best		**xxx**

ANSWER FRAME

"What is capital rationing? How are projects selected under it?"

1. Meaning: funds are restricted, and management has to choose from amongst available alternative investments, all with positive NPVs.
2. Objective: maximise the total NPV obtainable within the funds available (wealth maximisation).
3. Independent and divisible projects: modify the NPV rule; rank by NPV per rupee of capital (equivalently PI); accept in order, the last project in part.
4. Indivisible projects: rank by absolute NPV, examine combinations within the limit, and select the combination with the highest total NPV.
5. Note PI's limitation for indivisible projects, and that negative-NPV projects are never included.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: ranking divisible projects by the size of their NPV**
   Fix: a rationed budget is spent rupee by rupee; rank divisible projects by NPV per rupee.

2. **Taking indivisible projects in PI order and stopping**
   Fix: list the combinations that fit and compare their total NPVs; here P + Q + S gives ₹20.41 lakh.

3. **Spending the leftover on a negative-NPV project**
   Fix: a project with NPV < 0 (PI < 1) is rejected under rationing as it is without it.

4. **Taking the full NPV of a part project**
   Fix: NPV of the part = NPV × funds remaining ÷ investment.

RECAP

- Capital rationing: funds are restricted, so management chooses among positive-NPV projects; the aim is the highest total NPV within the budget.
- Independent, divisible projects: rank by NPV per rupee of capital (NPV ÷ investment), accept in order, take the last in part with the same fraction of its NPV.

- NPV per ₹1 = PI − 1 when the outlay is at year 0: the same ranking; PI is the better technique under capital rationing.
- Indivisible projects: rank by absolute NPV, examine combinations within the limit, select the highest total NPV; a lower-ranked project may replace a higher-ranked one (Illustration 13: E + B + D beats E + B + C).
- PI fails as a guide where projects are indivisible (SM 7.32).
- Negative-NPV projects are never accepted, even with idle funds; unused budget is not a reason to accept.
- (Loose sweets go in by value per kilo; pressure cookers go in by trying the sets.)

CHAPTER 8

Dividend Decision

8.1 Forms of dividend, buy-back and determinants of dividend policy

Key idea

Dividend is the part of profit after tax distributed to shareholders; the rest is retained for growth (g = b × r). Cash dividends and buy-backs pay out cash, bonus shares only capitalise reserves, and stock splits only change the number of shares. The right payout depends on investment opportunities, cost of capital, shareholders' expectations, legal and tax constraints, and the share price.

FORMATS AND FORMULAS

Reproduce these from memory.

Formula

Payout and retention

$$\text{Payout ratio} = \frac{\text{DPS}}{\text{EPS}} \qquad b = 1 - \text{Payout ratio}$$

DPS = dividend per share; b = retention ratio.

∑ **Formula**

Growth from retained earnings

$$g = b \times r$$

b = retention ratio, r = rate of return on investment (SM 8.8).

∑ **Formula**

Lintner's model

$$D_1 = D_0 + [(\text{EPS} \times \text{Target payout}) - D_0] \times A_f$$

Af = adjustment factor (speed of adjustment), between 0 and 1 (SM 8.14).

∑ **Formula**

Share buy-back

$$\text{Shares bought back} = \frac{\text{Amount}}{\text{Buy-back price}}$$

$$\text{BVPS after} = \frac{\text{Book value before} - \text{Amount paid}}{\text{Shares remaining}}$$

Cash paid out reduces net worth; with earnings unchanged, EPS after = earnings ÷ shares remaining (SM 8.54, problem 9).

Ex-dividend price = cum-dividend price − dividend per share (SM 8.4; 8.54, problem 8).

Definition

Stock split

Stock split means splitting one share into many, say, one share of ₹500 into 5 shares of ₹100. (SM 8.33)

Definition

Share buyback

Share buyback, in simple terms, means buying/repurchasing own shares by the company resulting into decrease in share capital of the company. Thus, the shares bought back are cancelled leading reduction in outstanding number of shares. (SM 8.33)

ANSWER FRAME

"State dividend decision. Explain the factors which govern it" (SM theory question 1; ICAI's answer points to paragraphs 1, 2 and 6 of the chapter): the dividend decision as one of the three long-term finance decisions, with the financing and investment decisions; meaning (part of PAT distributed; the rest retained for growth); then the nine determinants in ICAI's order, one line each: availability of funds, cost of capital, capital structure, stock price, investment opportunities, trend of industry, expectation of shareholders, legal constraints (Section 123), taxation.

"Explain the advantages and disadvantages of stock dividend" (SM theory question 2): advantages to

shareholders (three) and to the company (two), then limitations to shareholders and to the company, as in the tables above.

"Discuss the practical considerations in dividend policy" (SM theory question 3): (a) financial needs, with the mature and growth company contrast; (b) constraints: legal, liquidity, access to capital market, investment opportunities; (c) payout policies: (i) constant dividend policy (fixed amount, dividend equalisation reserve); (ii) stable dividend policy (constant payout ratio; dividends based on long-term sustainable earnings, Lintner's target payout).

"State the meaning of stock split; explain its advantages and disadvantages" (SM theory question 5; September 2025): ICAI's definition with the ₹500 into 5 × ₹100 example; two advantages; two limitations.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the stock split that "distributed profits"**
   Fix: bonus shares capitalise reserves and are a form of dividend; a stock split only divides each share into more shares of lower face value.

2. **Using the market price instead of the buy-back price**
   Fix: in Tapri's tender offer, ₹96 crore buys 16 lakh shares at ₹600, not 20 lakh at the market price of ₹480.

3. **Lintner without the brake**
   Fix: only the fraction Af of the gap is closed: ₹8 + (₹12 − ₹8) × 0.5 = ₹10, not ₹12.

4. **Payout in place of retention in g = br**
   Fix: b is the retention ratio, 1 − payout.

RECAP

- Dividend: the part of profit after tax distributed to shareholders; the rest is retained for growth, $g = b \times r$.
- Forms: cash dividend; share repurchase (treasury or cancelled shares), same effect on wealth as a cash dividend; stock dividend (bonus shares), which capitalises reserves and leaves net worth unchanged.

- Bonus conditions (SEBI): not in lieu of cash dividend; authorised by AOA, all shares fully paid; no default on loans, interest or statutory dues; only from share premium and free reserves.
- Stock split: one share into many of lower face value; makes shares affordable; costs money and may attract speculators. Not a distribution of profits.
- Buy-back: open market or tender offer; reduces shares outstanding; BVPS after = (book value − amount paid) ÷ shares remaining.
- Determinants: funds, cost of capital, capital structure, stock price, investment opportunities, industry trend, shareholders' expectations, legal constraints, taxation.
- Payout policies: constant dividend (fixed amount, dividend equalisation reserve); stable dividend (constant payout ratio, or dividends based on long-term sustainable earnings). Lintner: $D_1 = D_0 + [(EPS \times \text{target payout}) - D_0] \times Af$, two parameters.
- (A bonus issue cuts the pie into more slices: nobody gets more pie.)

8.2 Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

Key idea

Walter and Gordon: dividend policy is relevant; compare r with Ke. r > Ke: zero payout is optimum; r < Ke: 100%; r = Ke: no optimum (Walter: every payout is optimum). MM: dividend policy is irrelevant; value depends on earnings and investment policy. Graham and Dodd: P = m(D + E/3), so dividends weigh more than retained earnings.

FORMATS AND FORMULAS

Reproduce these from memory; write the formula, then the figures.

Formula

Graham and Dodd (traditional approach)

$$P = m\left(D + \frac{E}{3}\right)$$

m = a multiplier (SM 8.39).

∑ **Formula**

Walter's model

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

r = return on investment; Ke = capitalisation rate (SM 8.22).

∑ **Formula**

Gordon's model

$$P_0 = \frac{E_1(1-b)}{K_e - br} \qquad P_0 = \frac{D_0(1+g)}{K_e - g} = \frac{D_1}{K_e - g}$$

b = retention ratio; g = br; valid only if Ke > g (SM 8.27, 8.29).

∑ **Formula**

Zero growth and multi-stage DDM

$$P_0 = \frac{D}{K_e} \qquad P_0 = \sum_{t=1}^{n} \frac{D_t}{(1+K_e)^t} + \frac{D_{n+1}}{K_e - g} \times \frac{1}{(1+K_e)^n}$$

The second term is the price at the end of the high-growth years, discounted (SM 8.30; 8.51 and 8.54).

∑ **Formula**

Modigliani-Miller

$$P_0 = \frac{P_1 + D_1}{1 + K_e} \qquad nP_0 = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

Δn = funds required ÷ P_1, where funds required = I − (E − nD_1) (SM 8.17–8.18).

The MM four steps (SM 8.19–8.21, Illustration 1)

Step	Dividend paid	Dividend not paid
1. Year-end price	$P_1 = P_0(1 + Ke) - D_1$	$P_1 = P_0(1 + Ke)$
2. Funds required	$I - (E - nD_1)$	I − E
3. New shares	Δn = funds required ÷ P_1	Δn = funds required ÷ P_1
4. Value of firm	$[(n + \Delta n)P_1 - I + E] \div (1 + Ke)$	Same formula: same value

ANSWER FRAME

"MM hypothesis is based on certain assumptions. What are they?" (May 2026 Q4(a), 4 marks): the five, in ICAI's order and words: perfect capital markets; no taxes (with the reason: universal applicability, since tax rates differ between countries); fixed investment policy; no floatation or transaction cost; no risk of uncertainty, one discount rate for all securities and periods.

"List any four assumptions of Gordon's model" (September 2024 Q4(b), 4 marks): all-equity firm; IRR constant; Ke constant; retention ratio constant; growth rate g = br constant; Ke > g; investments financed by retained earnings only. Give the reason ICAI attaches to each.

"Briefly explain the assumptions of Walter's model" (old scheme May 2022 Q6(b), 4 marks): the six assumptions above.

"State the limitations of Walter's model" (MTP Series I for January 2026, 2 marks): does not consider all factors, and Ke is difficult to determine; ignores taxation, legal and contractual obligations, management policy and attitude.

Numerical answers: working notes first (EPS after preference dividend, DPS, r and Ke, including Ke from a P/E ratio), then the formula, the figures and the result; end with the r against Ke conclusion in one sentence.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: forgetting the preference dividend**
   Fix: deduct the preference dividend first: (72 − 18) ÷ 6 = ₹9.

2. **Last year's dividend in Gordon's numerator**
   Fix: the formula needs next year's dividend, $D_0(1 + g)$ = ₹12, so P = ₹240.

3. **MM with the wrong funds required**
   Fix: in Sheetal Cold Chain's Policy B, ₹56,00,000 − (₹70,00,000 − ₹60,00,000) = ₹46,00,000, at the dividend-case price of ₹125; I − E would even say ₹14,00,000 is left over.

4. **The optimum payout as a "balanced" 50%**
   Fix: compare r with Ke and write the corner; only r = Ke makes the payout irrelevant.

RECAP

- Relevance: Walter and Gordon; irrelevance: MM; Graham and Dodd weight dividends more: P = m(D + E/3).
- Walter: P = [D + (r ÷ Ke)(E − D)] ÷ Ke; r > Ke zero payout, r < Ke 100%, r = Ke every payout optimum.

- Gordon: $P_0 = E_1(1 - b) \div (Ke - br) = D_1 \div (Ke - g)$; needs $Ke > g$; $r = Ke$ has no optimum ratio.
- Bird-in-hand: risk-averse investors prefer certain current dividends. DDM: zero growth $D \div Ke$; constant growth; multi-stage = PV of dividends + PV of the price at the end of the high-growth phase.
- MM: value depends on earning power and investment policy; $P_0 = (P_1 + D_1) \div (1 + Ke)$; $nP_0 = [(n + \Delta n)P_1 - I + E] \div (1 + Ke)$; $\Delta n = [I - (E - nD_1)] \div P_1$.
- MM assumptions: perfect markets, no taxes, fixed investment policy, no floatation or transaction cost, no uncertainty; dividends and repurchases are equivalent.
- Walter limitations: not all factors, Ke hard to determine, ignores taxes, legal and contractual obligations, management attitude.
- Preference dividend before EPS; D_1, not D_0, in Gordon; $Ke = 1 \div P/E$ when only P/E is given. (Plant the seeds only if they beat the market.)

Interactive version: PassBook app, lesson Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

CHAPTER 9

Management of Working Capital

9.1 Working capital, the operating cycle and financing policy

Key idea

Working capital is the money stuck in the operating cycle: current assets minus current liabilities. The longer the cycle (days in raw material, work-in-progress, finished goods and debtors, less suppliers' credit), the more of it a business needs. Holding more is safer but earns less; holding less earns more but risks running dry. The permanent part is financed long-term, the temporary part short-term.

FORMATS AND FORMULAS

Reproduce the operating cycle formula and the five component formulas from memory: the denominators carry the marks.

Formula

Working capital

$$\text{Net working capital} = \text{Current assets} - \text{Current liabilities}$$

$$\text{Gross working capital} = \text{Current assets}$$

∑ **Formula**

Operating cycle and number of cycles

$$\text{Operating cycle} = R + W + F + D - C$$

$$\text{Number of cycles} = \frac{360 \text{ or } 365}{\text{Operating cycle}}$$

∑ **Formula**

The five components (SM 9.20)

$$R = \frac{\text{Average stock of raw material}}{\text{Average cost of raw material consumption per day}}$$

$$W = \frac{\text{Average work-in-progress inventory}}{\text{Average cost of production per day}}$$

$$F = \frac{\text{Average stock of finished goods}}{\text{Average cost of goods sold per day}}$$

$$D = \frac{\text{Average receivables}}{\text{Average credit sales per day}}$$

$$C = \frac{\text{Average payables}}{\text{Average credit purchases per day}}$$

Average = (opening + closing) ÷ 2. Raw material consumed = opening stock + purchases − closing stock. Work-in-progress inventory may also be termed works cost (SM 9.19).

Item	Numerator	Denominator (per day)
R	Average raw material stock	Raw material consumed
W	Average WIP	Cost of production
F	Average finished goods	Cost of goods sold
D	Average receivables	Credit sales
C	Average payables	Credit purchases

∑ **Formula**

The ratio tests of the optimum (SM 9.7)

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Current assets} - \text{Inventory} - \text{Prepaid expenses}}{\text{Current liabilities}}$$

Thumb rule: current ratio 2 for a manufacturing firm, quick ratio at least 1; the real optimum depends on the business.

Σ **Formula**

Investment policy measures (SM 9.15–9.16)

$$\frac{\text{Current assets}}{\text{Fixed assets}}$$

$$\text{Return on total assets} = \frac{\text{EBIT}}{\text{Total assets}}$$

Higher CA/FA ratio: conservative; lower: aggressive (fixed assets constant).

Definition

Permanent and temporary working capital (SM 9.5)

Permanent working capital refers to the base working capital, which is the minimum level of investment in the current assets that is carried by the entity at all times to carry its day-to-day activities. **Temporary working capital** refers to that part of total working capital which is required by an entity in addition to the permanent working capital; it is also called variable or fluctuating working capital.

ANSWER FRAME

"Discuss the factors to be considered while planning working capital requirement"

Name and explain the nine factors in ICAI's order, one line of example each: need for cash; desired level of inventory; receivables; short-term financing options; nature of business; market and demand conditions;

technology and manufacturing policies; operating efficiency; price level changes and exchange rate fluctuations. In a case ("a pharmacy chain facing rising costs and uncertain demand"), tie each factor to a fact of the case.

"Discuss the liquidity vs profitability issue"

1. Each rupee of capital bears a cost; liquidity must be maintained with that cost in mind.
2. Higher working capital may raise revenue and profit, but funds tied up in idle assets reduce liquidity and the opportunity to earn better returns.
3. Hence a trade-off is required that increases profitability without disturbing day-to-day functioning, through the 3 Es: economy in financing, efficiency in utilisation, effectiveness in achieving objectives.
4. The component table: inventory, receivables, prepaid expenses, cash, payables, with the higher side, the trade-off tool and the lower side.

"Permanent and fluctuating working capital are both necessary. Describe."

Define each in ICAI's words (base level carried at all times, financed long-term; the part in addition to it, due to fluctuations in sales, e.g. seasonal inventory); a sketch of the two diagrams (flat and rising permanent line with the temporary bulge above); conclude that both facilitate production and sales through the operating cycle, and

that the permanent part is financed by long-term and the temporary part by short-term sources.

"Explain the approaches of working capital investment"

Point	Aggressive	Conservative
Current assets	Minimal	High
Inventory, credit, cash	Low, strict, little	High, liberal, ample
Liquidity and risk	Poor liquidity, higher risk	Greater liquidity, lower risk
Return	Higher	Lower
Danger	Stock-outs, lost growth	Cost of capital, obsolescence, bad debts
CA ÷ FA	Lower	Higher

Add: moderate lies between and is what most firms follow.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the samosas measured against the wrong denominator**
   Fix: each stage over the flow that leaves it (the trick above); raw material consumed = opening + purchases − closing; only credit sales and credit purchases for D and C.

2. **The creditors that were added**
   Fix: R + W + F + D − C, and check: the net cycle is always shorter than the gross cycle.

3. **"The higher the current ratio, the better"**
   Fix: in theory answers say "neither too high nor too low"; the thumb rule of 2 (and quick ratio of 1) depends on the business.

4. **Swapping aggressive and conservative**
   Fix: aggressive takes risk to save funds; conservative pays for safety.

RECAP

- Net working capital = current assets − current liabilities; gross working capital = current assets (SM 9.3–9.4).
- Permanent: the base carried at all times, financed long-term; temporary (variable, fluctuating): the extra due to sales fluctuations, financed short-term.

- Too much: idle funds, low return, over-capitalisation. Too little: insolvency risk, lost sales. Thumb rule: current ratio 2, quick ratio 1.
- Nine determinants: cash, inventory, receivables, short-term financing, nature of business, market and demand, technology and manufacturing policy, operating efficiency, price and exchange rates.
- Investment approaches: aggressive (minimal current assets, high return, high risk), conservative (the reverse), moderate; higher CA ÷ FA = more conservative.
- Operating cycle = R + W + F + D − C; number of cycles = 360 (or 365) ÷ operating cycle.
- Denominators: consumption, cost of production, cost of goods sold, credit sales, credit purchases.
- Hedging (matching) approach: finance each asset with funds of the same maturity. (Diwali cabins are hired, not bought.)

Interactive version: PassBook app, lesson Working capital, the operating cycle and financing policy

9.2 Estimating working capital requirement

Key idea

Each current asset = the cost flowing through it per period × the periods it is held; each current liability = the expense that flows into it per period × the periods it stays unpaid. Value each item at the cost actually paid in (for cash cost basis: no profit, no depreciation), take half the conversion cost for work-in-progress when it accrues evenly, subtract the liabilities and add the safety margin.

FORMATS AND FORMULAS

Reproduce ICAI's statement from memory: the headings, the order and "Safety Margin" carry marks.

Estimation of Working Capital Requirements (SM 9.24–9.25)	₹	₹
I. Current Assets:		
Inventories: Raw materials; Work-in-process; Finished goods	-- -	-- -
Receivables: Trade debtors; Bills	-- -	-- -
Prepaid expenses		-- -

Estimation of Working Capital Requirements (SM 9.24–9.25)	₹	₹
Minimum cash balance		-- -
Gross Working Capital		**-- -**
II. Current Liabilities:		
Trade payables; Bills payable	-- -	
Wages payable; Payables for overheads	-- -	-- -
III. Excess of current assets over current liabilities [I − II]		**-- -**
IV. Safety margin		-- -
V. Net Working Capital [III + IV]		**-- -**

Interactive version: PassBook app, lesson Estimating working capital requirement

∑ **Formula**

Operating cycle method

$$\text{Working capital} = \frac{\text{Annual operating cost}}{\text{Number of operating cycles}}$$

Cash cost basis: annual operating cost less depreciation.

∑ **Formula**

Current assets

$$\text{Item} = \frac{\text{Annual cost base}}{\text{12 months (or 365 days)}} \times \text{Holding period}$$

Item	Annual cost base (total cost basis)	Cash cost basis
Raw materials	Material consumed	Same
Work-in-progress	Material + share of conversion cost	Without depreciation
Finished goods	Cost of production	Cash cost of production
Debtors	Cost of sales or sales value	Cash cost of sales
Prepaid expenses	The expense paid in advance	Same
Trade payables	Credit purchases	Same
Wages, overheads payable	The expense paid in arrear	Same (no depreciation)

∑ **Formula**

Cash costs (Illustration 4, SM 9.30)

$$\text{Cash cost of production} = \text{Materials} + \text{Wages} + \text{Cash manufacturing expenses}$$

$$\text{Cash cost of sales} = \text{Cash cost of production} + \text{Administrative expenses} + \text{Selling expenses}$$

Adjust for opening and closing finished stock when they differ (TYK Problem 5, SM 9.139).

∑ **Formula**

Work-in-progress (material at the start, conversion accruing evenly)

$$\text{WIP} = \text{Material for the period in process} + \tfrac{1}{2} \times \text{Conversion cost for that period}$$

ANSWER FRAME

"Discuss the estimation of working capital need based on the operating cycle process" (SM 9.115, Q3)

1. The operating cycle is one of the most reliable methods of computing working capital (the others: ratio of sales, ratio of fixed investments); it estimates working capital from the average holding period of current assets, related to costs.
2. Operating cycle = R + W + F + D − C, each component defined, with its formula (average balance ÷ daily cost or sales).
3. Number of operating cycles = 360 (365) ÷ operating cycle; working capital = annual operating cost ÷ number of cycles.
4. Item by item: raw materials, WIP, finished goods, debtors (at cash cost, excluding profit and depreciation), cash; less trade payables, wages and overheads payable.

5. Add a safety margin; the shorter the cycle, the lower the requirement.

The numerical, three ways

Loses marks

A statement with bare figures: "Debtors 8,88,750; Creditors 3,75,000..."

No base, no period: an examiner cannot give method marks to a wrong number.

Passes

Each line shows base × period ("94,80,000 × 1.5/12"), but the cash sales are left inside debtors and depreciation inside the finished goods.

Two of the traps cost the marks.

Full marks

Working notes first (cash cost of production, WIP, cash cost of sales and the credit sales' share, the prepaid rent), then ICAI's statement with every base and period visible, safety margin last, assumptions stated.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the overdraft that was counted as cash**
   Fix: cash = ₹1,75,000 (SM 9.143); and do not deduct bank borrowing as a current liability.

2. **Profit smuggled into the debtors**
   Fix: build the cash cost of sales working note first, take the credit sales' share, and the debtors come from it.

3. **WIP charged with a full period of conversion**
   Fix: full material, half conversion: ₹7,92,000. Use the stated percentage when the question gives a degree of completion.

4. **WIP doubled in double shift**
   Fix: the quantity in process is unchanged, because work started in the first shift is completed in the second (SM 9.31); only its value per unit changes (a lower fixed wage per unit, a higher night-shift wage, cheaper material if there is a discount).

RECAP

- Three methods: current assets holding period (operating cycle), ratio of sales, ratio of fixed investments (SM 9.16–9.17).

- Operating cycle method: working capital = annual operating cost ÷ number of operating cycles; cash basis: depreciation out first.
- Each item = annual base ÷ 12 (52, 360, 365) × period held; current liabilities other than bank credit are deducted.
- WIP with material at the start and conversion accruing evenly: full material + half conversion for the period in process.
- Cash cost basis: finished goods at cash cost of production, debtors at cash cost of sales (no profit, no depreciation).
- ICAI's statement: current assets (gross working capital) − current liabilities = excess; + safety margin = net working capital.
- Double shift: WIP units unchanged; other stocks rise, though not always in proportion; every item valued at the new cost per unit (lower fixed cost per unit).
- Overdraft limits are not cash; a margin "inclusive of itself" divides. (Count the platform, not the trains.)

Interactive version: PassBook app, lesson Estimating working capital requirement

9.3 Management of receivables, credit policy, factoring and forfaiting

Key idea

Credit is worth giving only up to the point where the extra contribution from extra sales exceeds the extra bad debts, cash discount, collection cost and the opportunity cost of the money locked in receivables. Factoring and forfaiting turn receivables into cash: factoring is worth it when its effective cost is below the firm's cost of borrowing.

FORMATS AND FORMULAS

Reproduce ICAI's evaluation formats from memory: the line order is what the examiner checks.

Statement showing the evaluation of credit policies (total approach) (SM 9.73)

Particulars
A. Expected profit:
Present policy
Proposed policy I
Proposed policy II

Particulars

(a) Credit sales

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars

(b) Total cost other than bad debts: (i) variable costs; (ii) fixed costs

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars

(c) Bad debts

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars

(d) Cash discount

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars

(e) Expected net profit before tax (a – b – c – d)

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars

(f) Less: tax; (g) expected profit after tax

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars

B. Opportunity cost of investments in receivables

Present policy

...

Proposed policy I

...

Proposed policy II

...

Particulars
Net benefits (A – B)
Present policy
...
Proposed policy I
...
Proposed policy II
...

Incremental approach (SM 9.74–9.75): A. incremental expected profit [(a) incremental credit sales, less (b) incremental variable and fixed costs, (c) incremental bad debts, (d) incremental cash discount, (e) incremental expected profit, (f) tax, (g) after tax]; B. required return on incremental investments [(a) cost of credit sales, (b) collection period, (c) investment in receivables = a × b ÷ 365 (360), (d) incremental investment, (e) required rate of return, (f) required return = d × e]; **incremental net benefits (A – B)**.

Statement showing the evaluation of a factoring proposal (SM 9.83–9.84)

Particulars	₹
A. Annual savings (benefit) on taking factoring service	
Cost of credit administration saved	...
Bad debts avoided	...
Interest saved due to reduction in average collection period: cost of annual credit sales × rate of interest × (present − new collection period) ÷ 360	...
B. Annual cost of factoring to the firm	
Factoring commission: annual credit sales × % of commission	...
Interest charged by factor on advance: amount available for advance × collection period ÷ 360 × rate of interest (annualised)	...
C. Net annual benefits / cost of factoring (A − B)	...

∑ **Formula**

Effective cost of factoring (SM 9.84)

$$\text{Rate of effective cost} = \frac{\text{Net annual cost of factoring}}{\text{Advances to be paid}} \times 100$$

$$\text{or} \quad \frac{\text{Net annual cost}}{\text{Amount available for advance}} \times 100$$

Amount available for advance = receivables − factoring commission − factoring reserve; advances to be paid = amount available for advance − interest deducted by the factor. A year is taken as 360 days.

∑ **Formula**

Cash discount and average collection period

$$\text{Cash discount} = \text{Credit sales} \times \text{Share of customers taking the discount} \times \text{Rate}$$

Average collection period = the sum, over the groups of customers, of each group's share of credit sales × the days that group takes to pay. From a receivables turnover ratio: collection period in months = 12 ÷ turnover (Illustration 13, SM 9.80).

Formula

Expected rate of return (Illustration 12)

$$\text{Expected rate of return} = \frac{\text{Incremental expected profit}}{\text{Incremental investment}} \times 100$$

Definition

Factoring (SM 9.82, 9.106)

The outright sale of accounts receivables to a factor or financial agency; a continuous arrangement in which the factor purchases the client's trade debts, with or without recourse, exercises control over the credit extended to customers and administers the client's sales ledger.

Definition

Forfaiting (SM 9.86)

An arrangement of bill discounting in which a financial institution or bank buys the trade bills or receivables from exporters, the exporter relinquishing the right to receive payment from the importer, providing immediate finance on a without recourse basis.

"Explain the three aspects of management of receivables"

1. **Credit policy**: credit standards, credit terms and collection efforts; the trade-off between profit on additional sales and the cost of carrying debtors and bad debts; credit period ("net 50") and cash discount policy (rate, discount period, net credit period: "3/15 net 60").
2. **Credit analysis**: how risky it is to advance credit to a party; due diligence or reputation check.
3. **Control of receivables**: follow-up and a suitable collection policy; laying down policies and executing them.

"Describe factoring" / "Describe the salient features of forfaiting"

Factoring: definition (outright sale of receivables to a factor, who bears the risk of collection and services the accounts for a fee); with or without recourse; advantages (immediate cash, predictable cash flows, no impact on debt-equity, not a loan, efficient record keeping and collection); the evaluation (savings against commission and interest; accept if the effective cost is below the cost of borrowing).

Forfaiting: meaning ("relinquish a right"; bank buys exporters' receivables without recourse); the four

functions in order; the five features; Exim Bank's Buyer's Credit as the example.

"Explain the ageing schedule"

Receivables classified by age from the date of invoice (0–30, 31–60, 61–90, 91–120 days and more); compared with earlier schedules and with other firms; gives closer control over the quality of individual accounts; shows old accounts accumulating; supplements the average collection period. A two-date table makes the point.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the profit locked up in debtors**
   Fix: total cost (or variable cost, when only the contribution ratio is given), and say which; ICAI uses sales only when no cost is given (TYK Problem 11, SM 9.151).

2. **The discount rate applied to everybody**
   Fix: credit sales × share taking the discount × rate.

3. **Commission for one period compared with an annual saving**
   Fix: put everything on an annual basis: × 360 ÷ 72.

4. **Factoring and forfaiting treated as the same thing**
   Fix: forfaiting is always without recourse and for export receivables; factoring may be either way and domestic or international.

RECAP

- Objective: optimise the return on investment in receivables; three aspects: credit policy, credit analysis, control of receivables.
- "3/15 net 60": 3% discount within 15 days, otherwise pay within 60; cash discount = credit sales × share

taking it × rate.

- Costs of maintaining receivables: funds blocked (interest or opportunity cost), administration, collection, default.
- Total approach: expected profit − opportunity cost of investment in receivables; incremental approach: incremental profit − required return on incremental investment; both pick the same policy.
- Opportunity cost = total cost of credit sales × collection period ÷ 360 (365) × required rate; fixed cost = (average − variable cost per unit) × present units.
- Factoring: sale of receivables, with or without recourse, no impact on debt-equity; accept if effective cost < cost of borrowing.
- Effective cost of factoring = net annual cost ÷ advances to be paid (or ÷ amount available for advance) × 100.
- Forfaiting: exporters' receivables sold to a bank without recourse; Exim Bank's Buyer's Credit. Ageing schedule: receivables by age, compared over time. (Hand the diary to the agent only if his cut is cheaper than the bank.)

Interactive version: PassBook app, lesson Management of receivables, credit policy, factoring and forfaiting

9.4 Treasury and cash management

Key idea

Cash management is a balance between liquidity and cost: enough cash to meet every payment on time, no rupee left idle, and every surplus rupee parked where it is safe, short-term and quickly saleable.

FORMATS AND FORMULAS

Reproduce the cash budget format and the Baumol formula from memory.

ICAI's cash budget format (SM 9.43–9.44)

Heading: (name of company) Co. Ltd., Cash Budget, Period (...).

Particulars	Month 1	Month 2	... to Month 12
Receipts: 1. Opening balance			
2. Collection from debtors			
3. Cash sales			
4. Loans from banks			
5. Proceeds from issue of shares, debentures etc.			

Particulars	Month 1	Month 2	... to Month 12
6. Miscellaneous receipts			
7. Other items			
Total			
Payments: 1. Payments to creditors			
2. Wages			
3. Overheads (factory rent, administration, selling and distribution)			
4. Interest			
5. Dividend			
6. Corporate tax			
7. Capital expenditure			
8. Other items			
Total			
Closing balance [Surplus (+) / Shortfall (–)]			

The layout with a minimum balance (ICAI Illustration 6, SM 9.46)

When a minimum balance is kept and surpluses are invested or deficits financed at each month end, ICAI's solution adds these lines below the receipts and payments:

Line	How it is found
Total cash available (A)	Opening balance + receipts
Total payments (B)	All cash payments
Minimum cash balance desired	Given
Total cash needed (C)	B + minimum balance
Surplus − deficit (A − C)	Positive: invest; negative: liquidate or borrow
Investment / financing (D)	Investments in brackets; liquidations and borrowings positive
Closing cash balance	A + D − B, which equals the minimum

∑ **Formula**

Baumol: optimum cash balance (SM 9.60)

$$C = \sqrt{\frac{2U \times P}{S}}$$

C = optimum cash balance (the amount transferred each time); U = annual (or monthly) cash disbursement; P = fixed cost per transaction; S = opportunity cost of one rupee p.a. (or p.m.). U and S must be for the same period.

∑ **Formula**

What follows from C

$$\text{Number of transfers} = \frac{U}{C}$$

$$\text{Average cash balance} = \frac{C}{2}$$

$$\text{Total cost} = \frac{U}{C} \times P \qquad + \frac{C}{2} \times S$$

The first two are how ICAI's suggested answer to November 2022 Q1(a) (old scheme) worked; the interval between transfers is days in the year ÷ number of transfers. At the optimum the two costs are equal.

∑ **Formula**

Cash cycle and minimum operating cash (Illustration 11, SM 9.68)

$$\text{Cash cycle} = \text{Debtors period} + \text{Inventory period} - \text{Creditors period}$$

$$\text{Cash turnover} = \frac{360 \text{ days}}{\text{Cash cycle}}$$

$$\text{Minimum operating cash} = \frac{\text{Total annual operating outlay}}{\text{Cash turnover}}$$

∑ **Formula**

Is a faster collection system worth it? (SM 9.55)

$$\text{Funds released} = \text{Collections per day} \times \text{Days of float saved}$$

$$\text{Accept if: Funds released} \times \text{Rate of return} > \text{Added cost of the system}$$

Definition

Treasury management

Planning, organizing and controlling the funds and working capital of an enterprise in order to ensure best use of funds, maintain liquidity, reduce overall cost of funds and mitigate operational and financial risk. (SM 9.37)

Definition

Float

The periods that affect cash as it moves through the different stages of the collection process: billing float, mail float, cheque processing float and banking processing float. (SM 9.53–9.54)

Definition

Optimum cash balance (Baumol)

That level of cash where the carrying costs and transactions costs are the minimum. (SM 9.59)

ANSWER FRAME

"Explain briefly the functions of the treasury department" (4 marks)

The five tasks, in ICAI's order, one or two lines each: **cash management** (collections, payments, surplus funds in an investment portfolio), **currency management** (set off

intra-group indebtedness, match receipts and payments, forward contracts), **fund management** (plan and source short, medium and long-term needs; temporary investment of surplus; capital structure; forecast rates), **banking** (relationship and negotiations with bankers; bank loans, commercial paper), **corporate finance** (acquisitions and divestments; investor relations).

"Explain Baumol's model of cash management"

1. Optimum cash level: where carrying costs and transaction costs are the minimum.
2. Carrying cost = opportunity cost (interest foregone on marketable securities); transaction cost = cost of converting securities into cash (clerical, brokerage, registration).
3. Formula $C = \sqrt{2UP/S}$ with every symbol defined.
4. The diagram: holding cost rising, transaction cost falling, total cost U-shaped, optimum at its lowest point.
5. The four assumptions; the limitation (no fluctuation allowed), which Miller-Orr overcomes.
6. "With example": a three-line calculation.

"Discuss the Miller-Orr cash management model"

1. Net cash flow completely stochastic; a control limit model; decides the time and size of transfers between an investment account and the cash account.
2. Limits: h (upper), z (return point), zero (lower).

3. At h, invest h − z in marketable securities; at the lower limit, transfer from securities to cash; in between, no transactions.
4. Limits set by the fixed cost of securities transactions, the opportunity cost of holding cash and the degree of likely fluctuations.
5. More realistic than Baumol; the finance manager can set the limits to the firm's liquidity requirements.
6. A labelled sketch.

Short notes

- **Virtual banking**: meaning, the four advantages, the reason for its popularity (speed, convenience, round the clock access).
- **Kinds of float**: the four, one line each, then concentration banking and the lock box as the cures.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the monthly U with the annual S**
   Fix: put U and S on the same period: $\sqrt{2 \times 24{,}00{,}000 \times 150/0.12} = ₹77{,}460$ (or the monthly U with a monthly S of 1%, which gives the same).

2. **Borrowing to zero instead of to the minimum**
   Fix: finance to the minimum: ₹43 lakh. ICAI's layout (total cash needed = payments + minimum) makes the slip impossible.

3. **Depreciation paid in cash**
   Fix: a cash budget has only cash: ₹8 lakh. The same goes for provisions and write-offs.

4. **Calling the optimum balance the average balance**
   Fix: holding cost = (C ÷ 2) × S; transaction cost = (U ÷ C) × P.

RECAP

- Treasury management: planning, organising and controlling funds and working capital for best use of funds, liquidity, lower cost of funds and less risk; five tasks: cash, currency, fund management, banking, corporate finance.

- Cash management balances liquidity against cost; Keynes's three needs: transaction, speculative, precautionary.
- Cash budget: the most significant device for planning cash; with a minimum balance, total cash needed = payments + minimum, and the surplus or deficit is invested or financed (ICAI Illustration 6 layout). Only cash items; no depreciation.
- Four floats: billing, mail, cheque processing, banking processing. Concentration banking cuts the time from remittance to spendable funds; a lock box eliminates the cheque processing float but costs per cheque. Accept if the income on released funds exceeds the added cost.
- Baumol: $C = \sqrt{2UP/S}$; transfers U ÷ C; average balance C ÷ 2; at the optimum, holding cost = transaction cost. Assumes certain, uniform cash use.
- Miller-Orr: random flows; limits h, z and zero; invest h − z at the top, sell securities at the bottom, nothing in between.
- Minimum operating cash = annual operating outlay ÷ cash turnover, where cash turnover = 360 ÷ cash cycle.
- Surplus goes to securities chosen for safety, maturity and marketability. (Fill the wallet once a week, not once a year.)

Interactive version: PassBook app, lesson Treasury and cash management

9.5 Management of inventory

Key idea

Inventory is working capital on a shelf. Hold enough to avoid stock-outs and no more, and price every extra rupee and every extra day of stock at the cost of the funds that finance it.

FORMATS AND FORMULAS

Reproduce the EOQ formula and the cost comparison layout from memory.

Formula

Economic order quantity (CMA Study Material 2.24)

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

A = annual requirement (units); O = cost per order; C = carrying cost per unit per annum, which includes the cost of funds invested in the stock.

∑ **Formula**

Average stock and the money in it

$$\text{Average stock} = \frac{Q}{2} + \text{Safety stock}$$

$$\text{Investment} = \text{Average stock} \times \text{Unit cost}$$

Q = order quantity.

∑ **Formula**

Total annual cost of a stock policy

$$\text{Total cost} = \text{Purchase cost} + \frac{A}{Q} \times O + \left(\frac{Q}{2} + \text{Safety stock}\right) \times C$$

Q = order quantity. Compare policies on this total; the purchase cost matters whenever the policies change the price.

∑ **Formula**

Inventory holding periods (SM 9.20)

$$R = \frac{\text{Average raw material stock}}{\text{Raw material consumed per day}}$$

$$F = \frac{\text{Average finished goods stock}}{\text{Cost of goods sold per day}}$$

In ICAI's words: average stock of raw material ÷ average cost of raw material consumption per day; average stock of finished goods ÷ average cost of goods sold per day; W = average work-in-progress inventory ÷ average cost of production per day.

∑ **Formula**

Working capital released by fewer days of stock

$$\text{Funds released} = \text{Cost per day} \times \text{Days cut}$$

$$\text{Annual saving} = \text{Funds released} \times \text{Carrying rate}$$

Cost per day on the right base: material consumed per day for raw materials, cost of production per day for finished goods.

Comparison statement for two stock policies

Particulars	Policy I (₹)	Policy II (₹)
Purchase cost (units × price)	xxx	xxx
Ordering cost (number of orders × cost per order)	xxx	xxx
Carrying cost (average stock × carrying cost per unit)	xxx	xxx
Total annual cost	**xxx**	**xxx**

Definition

Inventory control (CIMA, as quoted by ICAI)

The function of ensuring that sufficient goods are retained in stock to meet all requirements without carrying unnecessarily large stocks. (CMA Study Material 2.21)

Definition

Economic order quantity

The size of an order for which total of ordering and carrying cost are minimum. (CMA Study Material 2.23)

ANSWER FRAME

"Explain the scope of inventory management" or

"What does inventory management cover?"

1. Inventories are a major element of working capital; their investment must be properly controlled; the objectives are similar to those of cash management.
2. The seven problems in ICAI's order: minimum and maximum levels; size of inventory; issues, receipts and inspection procedures; EOQ; storage; obsolescence; control over movement.
3. The aim: sufficient stock for all requirements without unnecessarily large stocks.

"Discuss the trade-off between profitability and liquidity in inventory"

Point	Higher inventory	Lower inventory
Effect	Fewer stock-outs; uninterrupted production and sales	Less capital required; cash released
Risk	Funds locked up; carrying cost; obsolescence	Stock-out and loss of goodwill
Tools	EOQ, JIT, stock levels: carry the optimum level	EOQ, JIT, stock levels: carry the optimum level

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the interest that was saved twice**
   Fix: the cost of funds is inside the carrying cost; count it once.

2. **The whole lot on the shelf all year**
   Fix: average stock = Q ÷ 2 + safety stock; the investment and the carrying cost follow from it.

3. **Finished goods at selling price**
   Fix: finished goods at the cost of production (cash cost where the question uses the cash cost basis), WIP at its cost to date (SM 9.22–9.23).

4. **The cheaper order that ignored the price**
   Fix: when a policy changes the price, the purchase cost is a relevant cost and enters the comparison.

RECAP

- Inventories are a major element of working capital; their objectives are similar to those of cash management (SM 9.69).
- Scope, in ICAI's order, is seven problems: minimum and maximum levels, size of inventory, issues, receipts and inspection, EOQ, storage, obsolescence and control over movement.

- Trade-off: more stock means fewer stock-outs (profitability), less stock means less capital but a risk of stock-out and loss of goodwill (liquidity), and EOQ and JIT find the optimum (SM 9.11).
- EOQ = √(2AO ÷ C); C includes the cost of funds invested in the stock.
- Average stock = Q ÷ 2 + safety stock; money invested = average stock × cost per unit.
- R, W and F are measured on cost bases; shifting stock faster frees cash.
- Compare stock policies on total cost (purchase + ordering + carrying), and count the interest on released funds only once.
- Inventory is ideally financed by supplier credit; banks lend against it as a cash credit. (Buy for the week, not for the wedding season.)

9.6 Management of payables

Key idea

Trade credit costs nothing only when no discount is lost. When there is a cash discount, convert it into an annual rate for the extra days of credit and compare it with the cost (or return) of your own funds: take the discount when forgoing it is dearer; otherwise pay on the last day of credit.

FORMATS AND FORMULAS

Reproduce both formulas and the comparison statement from memory.

Formula

Nominal annual cost of not taking the discount (SM 9.99)

$$\frac{d}{100-d} \times \frac{365 \text{ days}}{t}$$

d = size of the discount (for 6%, d = 6); t = days credit outstanding – discount period. Multiply by 100 for a percentage.

∑ **Formula**

Cost of lost cash discount, with compounding (SM 9.99)

$$\left(\frac{100}{100-d}\right)^{365/t} - 1$$

Always above the nominal cost. Use 360 for 365 when the question says so.

∑ **Formula**

Funds needed to take a discount

$$\text{Funds} = \text{Purchases} \times \left(1 - \frac{d}{100}\right) \times \frac{t}{365}$$

Funds = the average extra funds tied up; Purchases = annual purchases. Paying t days earlier on every invoice ties up t days of discounted purchases. Net benefit = discount earned − cost of these funds.

The refuse-or-accept statement (ICAI Illustration 17 layout, SM 9.100)

Particulars	Refuse discount (₹)	Accept discount (₹)
Payment to supplier	invoice value	invoice value × (1 – d%)
Less: return from investing the discounted amount between the discount day and the due date	(xxx)	—
Net cost	**xxx**	**xxx**

The lower net cost wins. The return is on the discounted amount (₹98 out of ₹100), for t days, at the rate the firm can earn (or saves on borrowing).

Definition

Trade credit

A spontaneous source of finance normally extended to the purchaser organisation by the sellers or service providers; credit obtained without completing much formality is open account trade credit. (SM 9.103)

Definition

t in the discount formulas

The reduction in the payment period in days, necessary to obtain the early discount, or days credit outstanding − discount period. (SM 9.99)

ANSWER FRAME

"Discuss the costs of availing trade credit" (4 marks)

Open with ICAI's sentence: "Normally it is considered that the trade credit does not carry any cost. However, it carries the following costs:" then the four, each with one line: **(i) Price** (forgone discount, a high implicit cost), **(ii) Loss of goodwill** (discrimination against delinquent customers when supplies are short), **(iii) Cost of managing** (administrative and accounting costs), **(iv) Conditions** (minimum order size or regular orders). If the question asks for both sides, add the three costs of not taking trade credit: inflation, interest, inconvenience.

"Show the difference between trade credit and bank overdraft" (RTP September 2025 Q9(b))

Point

Meaning

Trade credit

Credit extended by suppliers, allowing the firm to delay payment for goods or services

Bank overdraft

Customers may withdraw in excess of the credit balance in their current account

Point

Duration

Trade credit

Usually 15 to 90 days

Bank overdraft

Repayable on demand, but generally continued for long periods by annual renewal of limits

Point

Cost

Trade credit

A type of interest free loan; failure to use it has an interest cost

Bank overdraft

Interest charged on daily balances

Point

How it arises

Trade credit

Generated automatically in the course of business

Bank overdraft

Borrower applies to the bank; operated like cash credit and current accounts

Point

Size

Trade credit

Grows automatically with the volume of business

Bank overdraft

A fixed limit is granted within which the borrower may overdraw

"Explain the importance of trade credit and accruals, and their cost"

1. Credit from suppliers in the normal course of trade; a short-term source; small firms depend heavily on it.
2. Advantages: easy availability, flexibility, informality.
3. Not cost-free: the supplier's opportunity cost of funds in receivables is passed on as a higher price or as a cash discount not extended; the forgone discount is an implicit cost.
4. Accrued expenses (wages, salaries, taxes paid at the end of the period) are built-in and automatic, with no explicit or implicit cost (SM 9.103–9.104).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the 2% that was compared with 12%**

   Fix: annualise the discount over t days first, then compare.

2. **t taken as the whole credit period**
   Fix: t = days credit outstanding − discount period = 23; the first 7 days are free either way.

3. **d ÷ 100 instead of d ÷ (100 − d)**
   Fix: 1.5 ÷ 98.5, not 1.5 ÷ 100; and in the comparison statement, the return is on ₹4,92,500, not on ₹5,00,000.

4. **Stretching sold as a saving**
   Fix: a payment beyond the agreed period oversteps the credit; ICAI lists the loss of goodwill and supply discrimination as costs. If the discount is not taken, pay on the last day of credit, not later.

RECAP

- Trade creditor: a spontaneous short-term source arising from ordinary business; slow payment causes ill-feeling, disrupted supplies and a bad image (SM 9.98).
- Costs of availing trade credit: price (forgone discount), loss of goodwill, cost of managing, conditions.
- Costs of not taking trade credit: impact of inflation, interest, inconvenience.
- Nominal cost of not taking the discount = d ÷ (100 − d) × 365 ÷ t; cost of lost cash discount = (100 ÷ (100 − d))^(365/t) − 1; t = credit days − discount days.

- Take the discount when forgoing it costs more than the cost (or return) of your funds; otherwise pay on the last day of credit, not later.
- Refuse-or-accept: payment refused less the return on the discounted amount for t days, against the discounted payment.
- Never compare a period discount with an annual rate. (₹3 to keep ₹97 for six weeks is not cheap money.)

9.7 Financing of working capital and bank credit

Key idea

Finance permanent working capital from long-term sources and temporary working capital from short-term ones. Among short-term sources, compare the effective cost per rupee you can actually use, and weigh credit rating, feasibility, reliability and restrictions before choosing.

FORMATS AND FORMULAS

Reproduce the effective cost formulas and the comparison layout from memory.

Formula

Loan with a compensating (minimum) balance

$$\text{Effective cost} = \frac{\text{Interest rate}}{1 - b}$$

b = the minimum balance as a fraction of the loan, counted only if the firm would not otherwise keep it. SM 9.156: 16.5% ÷ (1 − 0.10) = 18.33%. Amount to borrow for a usable sum S: S ÷ (1 − b).

∑ **Formula**

Discounted bill or commercial paper

$$\text{Cost} = \frac{\text{Discount} + \text{Charges}}{\text{Net proceeds}} \times \frac{365}{n}$$

Cost = effective annual cost; n = days to maturity; net proceeds = face value − discount − charges. Never divide by the face value.

∑ **Formula**

Bank finance as a share of net current assets (SM case scenario)

$$\text{Bank finance} = 75\% \times \text{NCA}$$

$$\text{Margin} = \text{NCA} - \text{Bank finance}$$

NCA (net current assets) = current assets − current liabilities other than the bank borrowing. The percentage is the bank's stipulation in the question.

Comparison statement of short-term sources

Source	Effective annual cost
Bank loan with minimum balance	Rate ÷ (1 – b)
Bills discounting	Discount and charges ÷ net proceeds × 365 ÷ n
Commercial paper	(Face – net proceeds) ÷ net proceeds × 365 ÷ n
Trade credit (discount forgone)	d ÷ (100 – d) × 365 ÷ t

Rank the sources, choose the cheapest that is feasible, and comment on rating, reliability and restrictions.

Definition

Commercial paper

An unsecured promissory note issued by a firm to raise funds for a short period; an instrument that enables highly rated corporate borrowers to borrow short term. (SM 9.104)

Definition

Line of credit

A commitment by a bank to lend a certain amount of funds on demand, specifying the maximum amount. (SM 9.108)

Definition

Letter of credit

An arrangement by which the issuing bank, on the instructions of a customer or on its own behalf, undertakes to pay or accept or negotiate, or authorizes another bank to do so, against stipulated documents subject to compliance with specified terms and conditions. (SM 9.108)

Definition

Bank guarantee

A facility that commercial banks extend on behalf of their clients in favour of third parties, who are the beneficiaries of the guarantees. (SM 9.108)

ANSWER FRAME

"Describe the forms of bank credit" (4 marks, any four)

Write each form with ICAI's one-line description, in ICAI's order: cash credit (continuous facility within a sanctioned limit); bank overdraft (short-term, urgent needs, within limits, callable at short notice); bills discounting (seller discounts the accepted bill; discounting limit); bills acceptance (company draws on the bank, which accepts it); line of credit (commitment to lend on demand up to a maximum); letter of credit (bank undertakes to pay against stipulated documents); bank

guarantee (on behalf of the client in favour of a third-party beneficiary). Four well-explained forms earn the marks; ICAI's answer also accepted short-term loans.

"Explain commercial paper: its advantages and limitations"

1. Definition: an unsecured promissory note for a short period; for highly rated corporate borrowers; freely negotiable interest rate.
2. Maturity 7 days to less than 1 year; denominations of ₹5 lakh or multiples.
3. Advantages (a) to (e) in ICAI's order: unsecured, no restrictive conditions; rollover; tailored maturity; available in a tight money market; cheaper than bank loans.
4. Limitations: only highly rated firms; neither redeemable before nor extendable beyond maturity.

"Assessment of working capital by banks after 1997"

The four points of SM 9.106–9.107 in order: MPBF withdrawn (April 1997); banks' own Board-approved methods within prudential guidelines and exposure norms; RBI's directed credit and prohibition of credit still apply; MPBF instructions withdrawn and lending-discipline guidelines optional.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the 11% loan that cost 11%**
   Fix: 11% ÷ 0.85 = 12.94%, and borrow ₹2,35,29,412 to have ₹2 crore to use.

2. **Discount divided by the face value**
   Fix: divide by the net proceeds.

3. **MPBF written as today's rule**
   Fix: RBI withdrew MPBF in April 1997; banks use their own Board-approved methods. Use a percentage only when the question stipulates it.

4. **The CP that was repaid early**
   Fix: CP can neither be redeemed before maturity nor extended beyond it; plan the maturity to the need, with a bank limit as standby.

RECAP

- Permanent working capital comes from long-term sources (debt, equity), temporary working capital from short-term sources (SM 9.102).
- Spontaneous sources arise naturally (trade credit, bills payable, accrued expenses); negotiated sources are arranged with lenders.
- Choice of source: cost, impact on credit rating, feasibility, reliability, restrictions, matching of

maturities.

- Commercial paper: unsecured promissory note, highly rated firms, 7 days to less than 1 year, ₹5 lakh or multiples; cheaper than bank loans; cannot be redeemed early or extended.
- Tandon and Chore Committees set the old norms; RBI withdrew MPBF in April 1997; banks now use Board-approved methods within prudential guidelines and exposure norms.
- Seven forms of bank credit: cash credit, overdraft, bills discounting, bills acceptance, line of credit, letter of credit, bank guarantee.
- Effective cost = annual cost ÷ usable funds: a loan with a minimum balance costs rate ÷ (1 − balance %); a discounted bill or CP costs its charge ÷ net proceeds, annualised. (Four doors, four rents: read the small print.)